Point-by-Point Response to Reviewers

Manuscript ID: **BMEF-D-26-00169**

Journal: BME Frontiers (*A Science Partner Journal / AAAS*)

Title: Precision Synthesis of Multi-Tracer PET via VLM-Modulated Rectified Flow for MCI Stage Stratification

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Location references use the final combined manuscript reference copy; supplementary page numbers also identify the standalone Supplementary Materials. The clean main text has the same pagination before the references.

General Response

We thank the Editor and both reviewers for their detailed assessment of manuscript BMEF-D-26-00169. The revised manuscript distinguishes cross-sectional MCI stage discrimination from longitudinal prognosis, reports task-dependent classification findings and clinical baselines, and clarifies the extent of diagnosis-conditioned filling in the clinical inputs.

The main changes concern frozen BiomedCLIP and tracer-adapter definitions; subject-level two-fold synthesis and independent three-fold classification protocols; absolute historical and reconstructed image-quality evidence; normalized regional uptake statistics and FDR families; classification uncertainty, paired comparisons and clinical controls; measured one-step forward cost; and the limits of external and clinical interpretation. We explicitly distinguish the historical input assets from reconstructed generator experiments and do not claim clinical equivalence or molecular confirmation.

Below we reproduce and answer all 20 Reviewer 2 comments, all 15 Reviewer 3 major comments and all 21 Reviewer 3 minor comments individually. Each response is linked to the implemented final manuscript or supplementary locations. Suggested analyses that were not completed are identified and are not represented as completed experiments. Editorial forms and submission-system requirements are tracked separately from scientific revisions.

Response to Editorial Requirements

EDITOR-01

Comment EDITOR-01:

- A cover letter with a list of changes or a rebuttal against each point raised in the review **Response:**

We have prepared the point-by-point response covering all 56 reviewer comments and a separate revision cover letter. The manuscript changes are identified by their actual final locations below.

EDITOR-02

Comment EDITOR-02:

- A separate revised manuscript file with highlighted changes **Response:**

A clean revised manuscript and a separate highlighted revision are included in the prepared delivery package. The highlighted version is derived from the preserved pre-integration manuscript, rather than from a different historical draft.

EDITOR-03

Comment EDITOR-03:

- Separate high-resolution figure files in editable file format. We prefer vector-based files but can accept the following file types: Adobe Illustrator (AI), Adobe Portable Document Format (PDF), Encapsulated PostScript (EPS), Photoshop (PSD), JPEG, PNG, and Tagged Image File Format (TIFF). Please see our figure guide for assistance in preparing your figures. **Response:**

The delivery package contains separate figure PDFs and high-resolution exports. Recreated schematic and quantitative figures include editable SVG and source code; existing composite images retain their supplied source files and documented limitations. We do not describe a raster-only panel as a fully editable vector reconstruction.

EDITOR-04

Comment EDITOR-04:

- Signed License to Publish and Conflict of Interest forms from all listed Authors. Forms can be found here: `https://www.editorialmanager.com/bmef/account/BMEF_LicensetoPublish.pdf`; `https://www.editorialmanager.com/bmef/account/BMEF_ConflictofInterest.pdf` **Response:**

The local integration has not verified signed License to Publish or Conflict of Interest forms from all authors. These forms must be supplied and checked by the authors before submission; no signature or completed-form status is inferred from the manuscript.

EDITOR-05

Comment EDITOR-05:

-For all images reused or obtained from other sources, provide documentation of your right to reuse and/or modify the image. Consult our guide for details:https://spj.science.org/pb-assets/SPJ/CustomPages/Forms/forauthors_permissions.docx **Response:**

No clearance of third-party reuse rights is asserted by the local integration. Recreated plots and the schematic have documented local sources; the authors must confirm any applicable permission requirements for reused material before submission.

EDITOR-06

Comment EDITOR-06:

-For any change in authorship, all authors must sign the Authorship Change form available here: https://www.editorialmanager.com/BMEF/account/Auth_Change_Form.pdf **Response:**

This integration did not change the author list or affiliations relative to the preserved manuscript baseline. If the submission record differs or an authorship change is requested, the required author confirmations and change form must be handled by the authors.

EDITOR-07

Comment EDITOR-07:

Please note that your revised manuscript will be sent out for re-review and BMEF (BME Frontiers) rarely allows multiple revisions. **Response:**

We acknowledge that the revised manuscript may undergo further external review and have addressed the reviewers point by point, with unsupported claims narrowed to the available evidence.

EDITOR-08

Comment EDITOR-08:

Your revision is due by Nov 04, 2026. **Response:**

We note that the decision letter states a revision deadline of 4 November 2026. The local preparation does not establish that an online submission has occurred or that an extension was granted.

EDITOR-09

Comment EDITOR-09:

To submit a revision, go to https://www.editorialmanager.com/bmef/ and log in as an Author. You will see a menu item call Submission Needing Revision. You will find your submission record there. **Response:**

The authors will upload the reviewed final files through the journal's Editorial Manager revision

workflow. This local integration has not logged into or submitted files to the journal system.

EDITOR-10

Comment EDITOR-10:

There is additional documentation related to this decision letter. To access the file(s), please click the link below. You may also login to the system and click the 'View Attachments' link in the Action column. **Response:**

The decision letter refers to additional attachments available through View Attachments. Their contents have not been verified in the local materials; the authors must inspect and account for them before submission.

Response to Reviewer 2

R2-1

Comment R2-1:

1. High scientific novelty: The integration of 3D rectified flow, a domain-adapted BiomedCLIP/VLM, and multi-tracer PET synthesis is a clear methodological contribution beyond conventional MRI-to-PET translation.

Response:

We sincerely thank the reviewer for the positive assessment of the methodological contribution of our work. We appreciate the recognition of the value of integrating 3D rectified flow, domain-adapted BiomedCLIP/VLM, and multi-tracer PET synthesis.

Action in Manuscript:

We thank the reviewer for this positive assessment. No separate manuscript change was required for this comment.

R2-2

Comment R2-2:

2. Strong clinical motivation: Synthesizing AV45-PET and FDG-PET from routine MRI addresses important limitations related to PET cost, radiation exposure, and accessibility.

Response:

We thank the reviewer for highlighting the clinical relevance of our study. Addressing the challenges of PET acquisition costs, radiation exposure, and limited accessibility through MRI-based synthesis of AV45-PET and FDG-PET is a central motivation for our work.

Action in Manuscript:

We thank the reviewer for this positive assessment. No separate manuscript change was required for this comment.

R2-3

Comment R2-3:

3. Cross-validation inconsistency: The Results section reports 3-fold cross-validation, whereas the preprocessing/methodology section reports a two-fold scheme. The exact protocol must be reconciled and stated consistently for each experiment.

Response:

We thank the reviewer for pointing out the insufficient clarification of the cross-validation protocols in the original manuscript.

We confirm that the two-fold and three-fold cross-validation schemes were employed for different experimental stages. Specifically, two-fold cross-validation was used for PET image synthesis, where the training and held-out subsets were exchanged between folds to generate out-of-fold (OOF) synthetic PET images for the study subjects. Subsequently, a separate three-fold cross-validation scheme was applied to the downstream diagnostic classification tasks using the generated synthetic PET images to evaluate the contribution of different imaging inputs to classification performance.

Therefore, the two cross-validation schemes are not contradictory but correspond to distinct evaluation stages: PET image synthesis and downstream diagnostic classification. We now clarify their respective purposes and consistently distinguish these protocols in the Results and Methods sections of the revised manuscript to avoid ambiguity.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / evaluation split (p. 18, lines 457–457).
- tab:reproducibility (Supplementary Materials, p. 9; combined reference copy p. 33).
- Results / generation evaluation scope (p. 5, lines 118–118).

Revised Text in Manuscript (Materials and Data Preprocessing / evaluation split):

"PET synthesis used subject-level two-fold cross-validation with fixed, approximately equal partitions (929 and 928 participants; split seed 42)."

R2-4

Comment R2-4:

4. Overstatement of prognostic stratification: EMCI-versus-LMCI classification is a stage-discrimination task and does not by itself establish longitudinal prognosis or future conversion to AD. The terminology should be moderated unless longitudinal conversion outcomes are analyzed.

Response:

We thank the reviewer for highlighting the important distinction between MCI stage discrimination and longitudinal prognostic prediction.

Our original motivation for relating EMCI/LMCI classification to disease progression was based on previous longitudinal studies showing that patients with LMCI generally exhibit more severe cognitive impairment and have a higher subsequent risk of conversion to dementia than those with EMCI (Aisen et al., 2015; Lin et al., 2022; Petersen, 2021). This clinical evidence motivated us to include EMCI-versus-LMCI classification as a downstream task to evaluate the performance of synthetic PET in distinguishing clinically relevant MCI stages.

Nevertheless, we agree that such population-level associations do not establish individual-level prognostic prediction. Our study evaluated cross-sectional classification performance without analyzing longitudinal conversion outcomes. Therefore, the EMCI-versus-LMCI classification results

cannot, by themselves, demonstrate the ability to predict future conversion to AD or disease progression.

To address this concern, we will explicitly describe this task as **MCI stage discrimination** and revise the relevant terminology and conclusions throughout the manuscript. In particular, we now remove unsupported claims regarding prognostic prediction and future disease progression, ensuring that our conclusions accurately reflect the scope of the analyses performed.

1. Aisen, P. S., et al. (2015). Alzheimer's Disease Neuroimaging Initiative 2 Clinical Core: Progress and plans. *Alzheimer's & Dementia*, 11(7), 734–739. https://doi.org/10.1016/j.jalz.2015.05.005 2. Lin, S.-Y., et al. (2022). The Clinical Course of Early and Late Mild Cognitive Impairment. *Frontiers in Neurology*, 13, 685636. https://doi.org/10.3389/fneur.2022.685636 3. Petersen, R. C. (2021). MCI Criteria in ADNI: Meeting Biological Expectations. *Neurology*, 97(12), 597–599. https://doi.org/10.1212/WNL.0000000000012588

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / diagnostic groups (p. 17, lines 413–413).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Abstract (p. 1, lines 18–18).
- Introduction / ill-posed inverse problem (p. 2, lines 54–54).
- Introduction / evaluation summary (p. 3, lines 76–77).
- Title (p. 1).
- PDF metadata.

Revised Text in Manuscript (Materials and Data Preprocessing / diagnostic groups):

"Participants were stratified into five clinically defined diagnostic categories to reflect the full continuum of Alzheimer's disease pathology: Cognitively Normal (CN; $n = 577$), Alzheimer's disease dementia (AD; $n = 317$), and three subtypes of Mild Cognitive Impairment (MCI)."

R2-5

Comment R2-5:

5. Potential clinical-information leakage: Age, sex, weight, and cognitive scores are embedded in the text conditioning. Because these variables are strongly associated with diagnosis, ablation experiments excluding cognitive scores are necessary to demonstrate that Syn-PET is not simply re-encoding diagnostic metadata.

Response:

We thank the reviewer for highlighting the potential overlap between cognitive scores and diagnostic labels. We acknowledge that this is a valid concern and therefore further investigated the influence of cognitive scores on the quality of synthetic PET images.

Specifically, we conducted an inference-time sensitivity analysis in which cognitive scores were removed while keeping the MRI input, tracer conditioning, and generator weights unchanged. Age, sex, and weight information were retained. This experiment included 1,856 subjects and 2,961 synthetic PET images. Compared with the full clinical conditioning, removing cognitive scores resulted in only minor increases in regional mean absolute error (ROI MAE) of approximately 0.000036 for FDG and 0.000079 for AV45, measured in normalized uptake units. The mean absolute voxel-wise differences between the synthetic images generated under the two conditions were approximately 0.001 for both tracers. Although some differences were statistically detectable, their absolute magnitudes were small, and the synthetic images maintained relatively stable reconstruction quality. These findings suggest that PET synthesis in DIReCT++ does not depend entirely on cognitive-score conditioning.

Our motivation for incorporating cognitive scores was to utilize clinically available complementary information to guide PET synthesis according to patients' cognitive characteristics, particularly considering the differences in cognitive impairment between finer MCI stages such as EMCI and LMCI. Nevertheless, we agree with the reviewer that the association between cognitive scores and diagnostic group membership warrants careful consideration when interpreting downstream classification results. Accordingly, we now further clarify the role of clinical conditioning in the revised manuscript and provide a more cautious discussion of the potential value of synthetic PET for disease diagnosis and MCI stage discrimination.

The fixed-weight score-removal experiment used the separate E1 Full checkpoint family. It cannot remove information already learned from diagnosis-conditioned filled inputs during training. In the completed classification analysis, Clinical-only and MRI+Clinical AUC exceeded MRI+synthetic dual PET in all three tasks; we do not claim independent incremental diagnostic value of the synthetic PET over those clinical fields.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- tab:clinical_missingness (Supplementary Materials, p. 8; combined reference copy p. 32).
- Materials and Data Preprocessing / clinical inputs (p. 18, lines 422–422).
- Results / framework conditioning (p. 3, lines 89–89).
- eq:composite_conditioning (p. 16, lines 359–359).
- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).
- Supplementary clinical score sensitivity and demographic stratification (Supplementary Materials, p. 26; combined reference copy p. 50).

Revised Text in Manuscript (Materials and Data Preprocessing / clinical inputs):

"In the prepared clinical inputs, mean Mini-Mental State Examination (MMSE) scores decreased from 29.40 in CN to 20.84 in AD; other clinical scores also varied across diagnostic groups (Table S1)."

R2-6

Comment R2-6:

6. Subject-level splitting must be explicit: Because ADNI contains longitudinal scans, the authors should confirm that all visits from a given participant remain within a single fold to avoid subject leakage between training and testing.

Response:

We thank the reviewer for highlighting the importance of subject-level data splitting and the potential risk of data leakage arising from longitudinal visits.

Although the original ADNI database contains longitudinal imaging data, the subset used in our study was selected such that only one imaging record was retained per subject. Consequently, no scans from different visits of the same subject could be assigned to different cross-validation folds, thereby preventing subject-level data leakage arising from repeated visits.

We now clarify this data selection criterion in the revised manuscript to avoid ambiguity.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / cohort (p. 17, lines 403–403).
- Materials and Data Preprocessing / evaluation split (p. 18, lines 457–457).

Revised Text in Manuscript (Materials and Data Preprocessing / cohort):

"This study leveraged neuroimaging data from 1,857 participants enrolled in the Alzheimer's Disease Neuroimaging Initiative (ADNI), spanning cohorts from ADNI-1 through ADNI-4."

R2-7

Comment R2-7:

7. External validation is incomplete for multi-tracer claims: OASIS-3 external evaluation includes AV45-PET but excludes FDG-PET. Therefore, external generalizability is demonstrated only partially and the multi-tracer generalization claim should be qualified.

Response:

We thank the reviewer for highlighting the need to clarify the scope of our external validation.

In this study, both AV45-PET and FDG-PET synthesis were evaluated internally on the ADNI dataset, whereas external validation on the independent OASIS-3 dataset was conducted only for AV45-PET. As described in the original manuscript (Section 4.3), FDG-PET scans in OASIS-3 were not included in the external evaluation due to the lack of a standardized PUP processing pipeline.

We agree that the current OASIS-3 results provide evidence of cross-dataset generalizability for AV45-PET synthesis only and cannot be directly extended to FDG-PET synthesis. Accordingly, we now clarify the scope of external validation and refine the relevant statements regarding multi-tracer generalizability in the revised manuscript to ensure that our conclusions accurately reflect the evaluations performed.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / OASIS (p. 18, lines 433–433).
- OASIS cohort description (Supplementary Materials, p. 1; combined reference copy p. 24).
- Results / generation evaluation scope (p. 5, lines 118–118).
- Results / external AV45 synthesis (p. 5, lines 141–141).
- fig:2 caption (p. 7).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Abstract (p. 1, lines 18–18).
- Introduction / evaluation summary (p. 3, lines 76–77).
- Results: PET synthesis (p. 4).

Revised Text in Manuscript (Materials and Data Preprocessing / OASIS):

"External evaluation used the Open Access Series of Imaging Studies (OASIS-3), a longitudinal neuroimaging and clinical cohort focused on normal aging and Alzheimer's disease (`https://www.oasis-brains.org`)."

R2-8

Comment R2-8:

8. Non-significance does not prove equivalence: Using $p > 0.05$ between real and synthetic PET as evidence of high fidelity is statistically insufficient. Equivalence testing (e.g., TOST) with clinically meaningful equivalence margins is more appropriate.

Response:

We thank the reviewer for pointing out that the absence of a statistically significant difference does not constitute evidence of equivalence. We agree that relying solely on paired comparisons with $p > 0.05$ between real and synthetic PET images is statistically insufficient to support claims of high fidelity.

To provide a more comprehensive evaluation of regional uptake agreement, we performed additional statistical analyses using the originally preserved subject-level paired data. In addition to paired t-tests and multiple-comparison correction, **we calculated the mean paired differences and their 95% confidence intervals, mean absolute error (MAE), Lin's concordance correlation coefficient (CCC), intraclass correlation coefficient (ICC), and Bland–Altman bias and limits of agreement**. Together, these metrics provide a more comprehensive assessment of mean bias, regional errors, and subject-level agreement.

The additional analyses showed that, across the 98 brain regions evaluated for each tracer, the median regional CCC values were 0.9745 and 0.9450 for AV45 and FDG, respectively, while the corresponding median regional MAE values were 0.0160 and 0.0159. Under the predefined multiple-comparison

correction procedure, no statistically significant mean paired differences were detected in the vast majority of regions and diagnostic groups.

Regarding the suggested two one-sided tests (TOST), we acknowledge the value of formal equivalence testing. However, the regional uptake values in our study were derived from per-image min–max intensity normalization rather than clinically standardized quantitative measurements such as SUV or SUVR. **Currently, a well-established equivalence margin with a clear clinical interpretation is lacking for measurements on this normalized intensity scale**. Defining an arbitrary equivalence margin without sufficient justification could lead to conclusions that lack clinical interpretability. Therefore, we did not perform TOST to claim clinical equivalence.

Accordingly, we now revise the statistical interpretation in the manuscript to clearly distinguish the absence of statistically significant mean differences, regional uptake agreement, and formal equivalence. We will also moderate the original statements that inferred high fidelity from $p > 0.05$. Furthermore, we will explicitly acknowledge in the Limitations section that our current analyses support agreement in normalized regional uptake values but do not establish clinical quantitative equivalence. Future work will explore PET quantitative evaluation scales with clear biological and clinical interpretations, the definition of appropriate equivalence margins, and corresponding formal equivalence testing methods.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 6, lines 149–149).
- Results: regional uptake analysis (p. 6, lines 157–157).
- Results: regional uptake analysis (p. 6, lines 163–163).
- Methods: regional statistical analysis (p. 20, lines 529–529).
- Methods: regional statistical analysis (p. 21, lines 547–547).
- tab:regional_agreement; tab:regional_effect_sizes (Supplementary Materials, p. 11; combined reference copy p. 35).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).

Revised Text in Manuscript (Results: regional uptake analysis):

"To assess regional uptake patterns, the corresponding T1-weighted MRI scans were parcellated using SynthSeg [1], and the resulting Desikan–Killiany and aseg ROI masks were applied to spatially aligned real and synthetic PET images."

R2-9

Comment R2-9:

9. Multiple-comparison reporting should be clearer: FDR correction is appropriate, but the statistical family, number of tests, and adjusted p/q-values should be reported explicitly.

Response:

We thank the reviewer for the valuable suggestion regarding the transparency of multiple-comparison correction reporting. We agree that, although the original manuscript stated that false discovery rate (FDR) correction was applied, the statistical families, number of tests, and adjusted p/q-values were not sufficiently specified.

To address this concern, we have confirmed the original regional statistical data and performed a systematic re-evaluation of the regional comparisons between real and synthetic PET images.

For the paired regional uptake agreement analysis between real and synthetic PET, we performed two-sided paired t-tests across all 98 regions of interest (ROIs) for each tracer, both in the overall cohorts and within individual diagnostic subgroups. Benjamini–Hochberg FDR correction was applied separately within each statistical family, comprising 12 families and 1,176 regional comparisons in total. The corresponding results for representative brain regions in the overall cohorts, including raw p-values and FDR-adjusted q-values, have been compiled in a new supplementary table.

For the AD versus CN comparisons shown in Figure 4C–F, we additionally performed two-sided independent-samples t-tests separately for AV45 Real, AV45 Syn, FDG Real, and FDG Syn PET. Benjamini–Hochberg FDR correction was applied across all 98 ROIs within each of these four statistical families, comprising 392 comparisons in total.

The re-evaluation showed that all 20 significant annotations in Figure 4C/D remained statistically significant after FDR correction, while all 20 non-significant annotations in Figure 4E/F remained non-significant. Sensitivity analyses using Welch's t-test did not change the significance classifications of these regions. Thus, the existing significant and non-significant classifications in Figure 4C–F are supported by the FDR-adjusted results and do not require changes.

In the revised manuscript, we will explicitly report the statistical families and corresponding numbers of tests, clarify the FDR correction procedure and significance thresholds in the Statistical Analysis section and the relevant descriptions of Figure 4, and provide adjusted q-values for representative brain regions to improve the transparency and reproducibility of the statistical analysis.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 9, lines 179–179).
- Methods: regional statistical analysis (p. 21, lines 540–540).
- Methods: regional statistical analysis (p. 21, lines 547–547).
- fig:4 caption (p. 10).
- tab:regional_agreement; tab:regional_effect_sizes (Supplementary Materials, p. 11; combined reference copy p. 35).

Revised Text in Manuscript (Results: regional uptake analysis):

"Critically, independent-sample t-tests demonstrated that the significant differences between diagnostic groups observed in real PET data were also observed in the synthetic images (Fig. 4C, D; figs. S5 and S6)."

R2-10

Comment R2-10:

10. PET agreement metrics should be strengthened: SSIM, PSNR, MSE, and MAE are useful image-similarity metrics but are insufficient for biomarker agreement. ICC/CCC, Bland-Altman analysis, regional SUV bias, and calibration analyses would improve clinical validation.

Response:

We thank the reviewer for the valuable suggestion to strengthen the evaluation of PET biomarker agreement. We agree that image-similarity metrics such as SSIM, PSNR, MSE, and MAE alone are insufficient to characterize subject-level regional uptake agreement and systematic differences between synthetic and real PET images.

To address this concern, we performed additional analyses using the original subject-level paired data across all 98 regions of interest (ROIs) for each tracer. Specifically, we calculated Lin's concordance correlation coefficient (CCC), the intraclass correlation coefficient (ICC(A,1)), mean paired differences with 95% confidence intervals, and Bland–Altman bias and 95% limits of agreement. These complementary metrics provide a more comprehensive assessment of regional errors, subject-level agreement, and systematic differences. Across the 98 ROIs, the median regional CCC values were 0.9745 for AV45 and 0.9450 for FDG.

Furthermore, to examine the relationship between synthetic and real regional uptake values, we conducted subject-level ordinary least-squares regression analyses for the 16 representative tracer–ROI combinations shown in Figure 4A/B. Real PET regional uptake was used as the independent variable and synthetic PET regional uptake as the dependent variable. Regression slopes, intercepts, and their corresponding 95% confidence intervals were calculated.

The regression slopes ranged from 0.9794 to 1.0127 (median: 1.0007) for AV45 and from 0.9844 to 1.0101 (median: 1.0001) for FDG. These representative regions did not exhibit a consistent pattern of linear compression or expansion, although modest deviations from a unit slope were observed in some regions. The regression results provide additional information on how synthetic regional uptake varies with real uptake levels and complement the CCC, ICC, and Bland–Altman analyses.

Importantly, the regional uptake measurements in this study were derived from per-image min–max normalized PET intensities rather than clinically standardized quantitative SUV or SUVR measurements. Therefore, the reported agreement and regression results characterize normalized regional uptake only and should not be interpreted as evidence of clinical SUV/SUVR agreement, absolute quantitative accuracy, or clinically validated calibration. We will explicitly clarify this distinction and its implications in the revised manuscript. Future work will investigate PET synthesis and validation approaches that preserve clinically meaningful quantitative uptake measurements.

In the revised manuscript, we will expand the regional agreement supplementary table to include regression slopes, intercepts, and their 95% confidence intervals for the 16 representative tracer–ROI combinations, alongside the existing agreement and bias metrics. We now also clarify the corresponding statistical methods and limitations.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 6, lines 163–163).

- Methods: regional statistical analysis (p. 20, lines 529–529).
- fig:3 caption (p. 8).
- tab:regional_agreement; tab:regional_effect_sizes (Supplementary Materials, p. 11; combined reference copy p. 35).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).

Revised Text in Manuscript (Results: regional uptake analysis):

"Paired-sample t-tests between synthetic and real images revealed no statistically significant differences in normalized regional uptake across the principal brain regions shown in Fig. 4A, B for the overall cohorts of both tracers ($p > 0.05$)."

R2-11

Comment R2-11:

11. Classification uncertainty is underreported: ACC, AUC, sensitivity, and specificity should be accompanied by fold-wise mean ± SD or 95% confidence intervals rather than isolated point estimates.

Response:

We thank the reviewer for highlighting the insufficient reporting of uncertainty in the classification results. We agree that isolated performance estimates do not adequately characterize variability across cross-validation folds and that measures of dispersion and confidence intervals are necessary for a more comprehensive evaluation.

For the three classification tasks—AD versus CN, MCI versus CN, and EMCI versus LMCI—we evaluated classification performance using subject-level three-fold cross-validation and summarized the predictions from the held-out test sets. For each input configuration, we report the mean and standard deviation across folds for accuracy (ACC), area under the receiver operating characteristic curve (AUC), sensitivity (SEN), specificity (SPE), and balanced accuracy. We additionally provide 95% bootstrap confidence intervals based on subject-level resampling.

Furthermore, we provide ROC curves with 95% bootstrap confidence bands to illustrate classification performance and its uncertainty across the three tasks. The corresponding quantitative results and visualizations are presented in the main manuscript and Supplementary Materials.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).
- Supplementary classification tables and ROC (Supplementary Materials, p. 17; combined reference copy p. 41).

R2-12

Comment R2-12:

12. Claims of statistical superiority require direct tests: Statements such as "significantly outperforming" should be supported by formal paired comparisons, such as DeLong testing for AUC or paired bootstrap/McNemar tests where appropriate.

Response:

We thank the reviewer for emphasizing the importance of direct statistical comparisons when evaluating classification performance. We agree that differences in point estimates alone are insufficient to fully characterize performance differences between methods. Accordingly, we conducted paired analyses using predictions from the same held-out subjects. Stratified paired bootstrap resampling was used to estimate performance differences and their 95% confidence intervals (CIs), and fold-wise paired DeLong tests were performed for AUC comparisons, with Holm correction for multiple comparisons.

Specifically, for AD versus CN classification, combining MRI with synthetic dual-tracer PET improved AUC by 6.80 percentage points (95% CI: 3.39–10.22) and balanced accuracy by 9.06 percentage points (95% CI: 4.68–13.46) compared with MRI alone. For MCI versus CN classification, the AUC difference was −2.13 percentage points (95% CI: −6.30 to 2.10), without a clear improvement. For the more fine-grained EMCI versus LMCI classification task, AUC increased by 3.52 percentage points (95% CI: −2.12 to 9.50), while balanced accuracy improved by 6.91 percentage points (95% CI: 2.02–11.98). The fold-wise DeLong comparisons did not reach statistical significance after Holm correction.

The differences across classification tasks may be related to the underlying disease stages and their imaging characteristics. AD and CN represent relatively distinct clinical states, often accompanied by more pronounced group-level differences in cerebral metabolism and amyloid deposition, making their discrimination comparatively straightforward. In contrast, MCI is clinically and pathologically heterogeneous, with imaging characteristics that may overlap with those of cognitively normal individuals and patients with AD. Distinguishing EMCI from LMCI is particularly challenging because the differences in disease-related imaging features between these stages may be relatively subtle.

Notably, even in the challenging EMCI versus LMCI task, incorporating synthetic PET yielded higher balanced accuracy, suggesting that the synthesized images may provide useful information for fine-grained MCI stage discrimination. In the revised manuscript, we will present these findings alongside the corresponding confidence intervals and statistical comparisons, and refine the original statements regarding statistical superiority to provide a more accurate interpretation of the contribution of synthetic PET to downstream classification.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).
- Supplementary classification tables and ROC (Supplementary Materials, p. 17; combined reference copy p. 41).

R2-13

Comment R2-13:

13. One-step sampling terminology requires correction: The phrase "exact one-step sampling" is too strong for a distilled end-point predictor. The manuscript should distinguish exact ODE integration from learned one-step approximation.

Response:

We thank the reviewer for pointing out the imprecision in our description of one-step sampling. We agree that the continuous ODE process underlying rectified flow should be distinguished from the learned one-step approximation employed in DIReCT++.

Specifically, DIReCT++ generates the target PET image directly from the input MRI in a single forward pass, which does not imply exact integration of the underlying ODE. Accordingly, we now revise the expression "exact one-step sampling" to more accurately describe the learned one-step generation process.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / framework inference (p. 3, lines 106–106).
- fig:1 caption (p. 4).
- eq:onestep_generation / learning schedule (p. 17, lines 391–392).
- Results / framework opening (p. 3, lines 84–84).

Revised Text in Manuscript (Results / framework inference):

"At inference, the shared network predicts a velocity field for the selected tracer from MRI and the corresponding two-token text context."

R2-14

Comment R2-14:

14. Domain-adapted VLM details are insufficient: The manuscript should report embedding dimensionality, PET image-feature extraction, prompt construction, affine-adapter training, threshold τ selection, and optimization details to ensure reproducibility.

Response:

We thank the reviewer for pointing out the need for a more detailed description of the domain-adapted vision-language model. We agree that explicitly reporting the feature dimensionality, 3D PET feature extraction procedure, text conditioning construction, and adapter optimization process is important for improving the transparency and reproducibility of DIReCT++. We provide the following clarifications regarding these implementation details.

First, our study employs a pre-trained BiomedCLIP model to extract text and PET image features. Both the text and image encoders remain frozen, and their projected feature representations are

512-dimensional. We construct the conditioning information using separate tracer-specific text descriptions and patient-specific clinical descriptions. The clinical prompt includes age, sex, weight, and five clinical assessments: CDR, MMSE, GDS, FAQ, and NPI-Q. After BiomedCLIP encoding, each patient is represented by a 512-dimensional clinical feature vector. During generation, this vector is combined with the corresponding 512-dimensional target tracer feature vector to form two conditioning tokens, which are incorporated into the 3D generative network through cross-attention.

Regarding PET image feature extraction, we clarify that the 2D BiomedCLIP encoder is not applied directly to an entire 3D PET volume. In the adapter feature-extraction procedure, 160 two-dimensional slices are first extracted from each preprocessed PET volume along a specified array dimension. Each slice undergoes individual intensity normalization, resizing, and conversion to a three-channel image before being processed by the frozen BiomedCLIP image encoder, producing a 512-dimensional feature vector. The resulting slice-level features are then aggregated through average pooling to obtain a subject-level PET representation, which is used for adapting the tracer-specific text features. The revised implementation applies the same feature-extraction procedure to both FDG and AV45.

Regarding the affine-adapter architecture, each tracer is associated with a lightweight affine transformation consisting of one learnable scalar scaling parameter and one 512-dimensional bias vector. Accordingly, each adapter contains 513 trainable parameters, resulting in 1,026 trainable parameters across the two adapters. Throughout this adaptation procedure, the BiomedCLIP encoders and their projection layers remain frozen, and only the parameters of the two adapters are optimized, rather than performing end-to-end fine-tuning of the BiomedCLIP backbone.

During optimization, the tracer text embeddings and PET image features are first normalized. Cosine similarity is then used to measure the alignment between the adapted tracer embeddings and their corresponding PET features. The alignment terms are calculated by averaging the feature similarities across the training subjects for FDG and AV45, respectively. In addition, a cross-tracer similarity penalty is introduced to regulate the relative distance between the two tracer representations. The implemented objective employs a two-sided squared soft penalty rather than the one-sided hard constraint expressed in the original manuscript. The two adapters are jointly optimized using the Adam optimizer with a learning rate of 0.001 for 5,000 parameter updates.

For the similarity parameter τ, we use 0.82 as the target cross-tracer cosine similarity. This value corresponds to the two-decimal approximation of the cosine similarity between the original FDG and AV45 text embeddings before adaptation (0.8205). In this revision, we further provide a representation-level sensitivity analysis of τ, comparing different target values and a configuration without the cross-tracer similarity penalty.

We thank the reviewer again for this constructive suggestion. In the revised manuscript, we will supplement the descriptions of feature dimensionality, PET slice-based encoding, clinical prompt composition, adapter parameters, and optimization settings to improve the reproducibility of the model implementation and the different experimental configurations.

For clarity, the revised manuscript distinguishes the affine-adapter formulation and representation-level analyses from the reconstructed Full generator, which used original fixed tracer-text vectors. The BiomedCLIP backbone is frozen in both cases. We do not attribute the reconstructed Full generator performance to affine adaptation or claim a controlled generator-level frozen-versus-adapted gain.

Relevant references: [2].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / framework conditioning (p. 3, lines 89–89).
- Methods / DIReCT++ overview (p. 14, lines 318–318).
- Domain-Adapted VLM / feature extraction (p. 14, lines 326–326).
- eq:align_opt (p. 15, lines 338–338).
- eq:composite_conditioning (p. 16, lines 359–359).
- Supplementary conditioning methods and tab:supp_tau_adapter (Supplementary Materials, p. 10; combined reference copy p. 34).

Revised Text in Manuscript (Results / framework conditioning):

"Text guidance incorporates tracer descriptions and patient information (age, sex, weight and clinical scores) encoded by frozen BiomedCLIP encoders and injected into the RF model through cross-attention."

R2-15

Comment R2-15:

15. Equation (1) needs stronger justification: The semantic-similarity constraint between FDG and AV45 embeddings may conflict with the stated goal of preserving inter-tracer differences. A sensitivity analysis for τ and an explicit ablation are recommended.

Response:

We thank the reviewer for the valuable comments regarding the rationale and parameter sensitivity of the cross-tracer semantic similarity constraint. We agree that its design motivation and its role in balancing text–PET feature alignment and tracer-specific representations warrant further clarification.

In DIReCT++, we employ a pre-trained BiomedCLIP model to encode the textual descriptions of FDG and AV45, followed by two independent affine adapters that align the corresponding textual embeddings with their respective PET image features. However, optimizing only the individual text–PET alignment objectives may cause the two adapted guidance vectors to become highly similar**,** potentially diminishing the inter-tracer distinctions present in the original textual representations. We therefore introduce a cross-tracer similarity constraint, **not to maximize the similarity between the two representations, but to regulate their relative similarity during feature alignment**.

Regarding the selection of τ=0.82, we would like to clarify its specific basis. Before affine adaptation, the cosine similarity between the original FDG and AV45 textual embeddings produced by the pre-trained BiomedCLIP model was 0.8205374, which rounds to 0.82. Accordingly, we selected τ=0.82 as the target cross-tracer similarity, using the original relationship between the two textual representations as a reference. This allows the adapters to align the textual embeddings with their corresponding PET image features while maintaining approximately the relative similarity originally

encoded by BiomedCLIP. **Thus, τ=0.82 was not an arbitrarily assigned value, but was determined from the representational characteristics of the pre-trained text encoder**.

We also acknowledge that Equation (1) in the original manuscript presented a simplified inequality constraint to convey the underlying design principle of cross-tracer semantic consistency. In the actual adapter optimization, this principle was implemented using a differentiable squared penalty, `100 × (cos(z_FDG, z_AV45) − τ)²`, which encourages the cosine similarity between the adapted tracer embeddings to approach the specified target τ. This formulation penalizes deviations both above and below the target, thereby preventing the two tracer representations from becoming increasingly similar without constraint during text–PET alignment. We now clarify the implemented optimization objective and the rationale for selecting τ in the revised manuscript.

To further validate this design, we conducted additional adapter-level parameter sensitivity and cross-tracer constraint ablation experiments. Using identical two-fold data splits, original text and PET features, initialization procedures, and optimization settings, we evaluated τ values of 0.70, 0.82, 0.90, and 0.98, together with an additional configuration in which the cross-tracer similarity penalty was completely removed.

The results demonstrate that τ effectively regulates the relative similarity between the adapted tracer guidance vectors. As τ increased from 0.70 to 0.98, the mean cosine similarity between the adapted FDG and AV45 embeddings increased from 0.7032 to 0.9807. At the originally adopted τ=0.82, the resulting similarities were 0.8213 and 0.8215 for the two folds, respectively, closely matching the specified target. In contrast, removing the cross-tracer penalty increased these similarities to 0.9900 and 0.9894, indicating that optimizing text–PET alignment alone caused the two guidance vectors to become highly similar.

Further analysis showed that the individual text–PET feature alignment also increased with τ. At τ=0.82, the mean alignment similarities were 0.9687 for FDG and 0.9661 for AV45. Although removing the cross-tracer constraint further increased these alignment values, it simultaneously brought the cosine similarity between the two adapted tracer embeddings close to 0.99. These findings indicate that maximizing individual text–PET alignment does not necessarily preserve the original representational distinctions between tracers. Instead, the cross-tracer similarity term provides a controllable mechanism for balancing individual feature alignment and inter-tracer representational separation.

Therefore, the rationale for selecting τ=0.82 is to use the original similarity between the BiomedCLIP textual representations as a reference, maintaining substantial text–PET feature alignment while preventing the two adapted guidance vectors from becoming excessively similar.

The additional experiments directly demonstrate the influence of τ on adapter-level representation learning and the relative geometry of the tracer embeddings. Since these analyses focus on the representation level rather than retraining PET synthesis models under different τ settings, we present them as supporting evidence for the parameter selection and similarity constraint mechanism, rather than as an optimization study of PET synthesis performance.

We now incorporate the rationale for τ=0.82, the actual optimization formulation, and the additional sensitivity and ablation results into the revised manuscript to improve the methodological clarity and reproducibility of our approach.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / framework conditioning (p. 3, lines 89–89).
- eq:align_opt (p. 15, lines 338–338).
- Supplementary conditioning methods and tab:supp_tau_adapter (Supplementary Materials, p. 10; combined reference copy p. 34).

Revised Text in Manuscript (Results / framework conditioning):

"Text guidance incorporates tracer descriptions and patient information (age, sex, weight and clinical scores) encoded by frozen BiomedCLIP encoders and injected into the RF model through cross-attention."

R2-16

Comment R2-16:

16. Demographic confounding needs control: ADNI groups differ significantly in age, sex, and other variables that are also used in the conditioning prompt. Matched, adjusted, or subgroup analyses are needed to quantify confounding effects.

Response:

We thank the reviewer for highlighting the potential confounding effects of demographic factors. We acknowledge that age and sex distributions differ across diagnostic groups in ADNI, making it important to further evaluate model performance across demographic subgroups.

To address this concern, we conducted additional age- and sex-stratified analyses. Subjects were stratified by age ($<$75 and $\geq$75 years) and sex (female and male), separately. Using the existing subject-level three-fold cross-validation test predictions, we compared MRI-only, MRI combined with synthetic dual-tracer PET, and MRI combined with real dual-tracer PET, along with the other imaging configurations. Paired bootstrap resampling was used to estimate 95% confidence intervals.

For AD versus CN classification, combining MRI with synthetic dual-tracer PET improved balanced accuracy across all four demographic subgroups, with gains ranging from 7.45 to 10.97 percentage points relative to MRI alone. For the more challenging EMCI versus LMCI classification, the balanced accuracy differences were also positive across all four subgroups, including a 9.64-percentage-point gain among subjects aged $\geq$75 years (95% CI: 1.23–18.05). In contrast, the MCI versus CN task did not show consistent improvements, suggesting that the contribution of synthetic PET may vary across classification tasks and patient populations.

We appreciate the reviewer's suggestion, which has enabled a more comprehensive evaluation of model performance across age and sex subgroups. We now incorporate these additional results into the revised manuscript and provide a more cautious discussion of the potential influence of demographic characteristics on downstream classification and the clinical value of synthetic PET.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- tab:supp_adni (Supplementary Materials, p. 1; combined reference copy p. 31).
- Materials and Data Preprocessing / clinical inputs (p. 18, lines 422–422).

- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).
- Supplementary clinical score sensitivity and demographic stratification (Supplementary Materials, p. 26; combined reference copy p. 50).

Revised Text in Manuscript (Materials and Data Preprocessing / clinical inputs):

"In the prepared clinical inputs, mean Mini-Mental State Examination (MMSE) scores decreased from 29.40 in CN to 20.84 in AD; other clinical scores also varied across diagnostic groups (Table S1)."

R2-17

Comment R2-17:

17. Baseline adaptation and fairness should be documented: Some comparator methods originate from segmentation or MRI-to-CT synthesis. The authors should clearly describe how each baseline was adapted and confirm identical data splits, preprocessing, tuning effort, and training budgets.

Response:

We thank the reviewer for highlighting the importance of documenting baseline adaptations and ensuring fair experimental comparisons. We agree that clarifying the task-specific adaptations, input conditions, and experimental settings of different methods is essential for improving the transparency and interpretability of the comparisons.

In response, we further examined and documented the implementations and experimental configurations of the baseline methods. For example, SwinUNETR was adapted for 3D PET intensity regression using continuous image outputs and an MSE loss function. The single-tracer DIReCT models were trained separately for FDG and AV45 synthesis. In addition, we included a 3D regression U-Net using the same clinical variables to further evaluate different generative architectures under comparable input conditions.

In the supplementary experiments, the methods were evaluated using the same subject cohort, fixed data splits, and preprocessing pipeline, with consistent evaluation metrics, including SSIM, PSNR, MSE, and MAE. We also documented the training configurations and evaluation epochs of the different models to facilitate a more accurate interpretation of their performance.

We acknowledge that differences remain in network architectures, conditioning inputs, and training configurations. Therefore, these comparisons primarily assess the overall performance of different methods rather than constituting strictly controlled ablations in which all experimental variables are held constant.

In the revised manuscript, we now provide clearer descriptions of the task adaptations and experimental settings of the principal baselines and further clarify the scope of the comparisons, thereby improving the transparency, objectivity, and reproducibility of our experimental evaluation.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / generation introduction (p. 4, lines 112–112).

- Results / reconstructed generation evidence (p. 5, lines 120–120).
- fig:2 caption (p. 7).
- Competing Methods / scope (p. 19, lines 470–470).
- Competing Methods / details; Image-Quality Evaluation (p. 19, lines 492–492).
- Supplementary generation tables and cases (Supplementary Materials, p. 12; combined reference copy p. 36).

Revised Text in Manuscript (Results / generation introduction):

"PET synthesis was evaluated using structural similarity (SSIM), peak signal-to-noise ratio (PSNR), mean squared error (MSE) and mean absolute error (MAE)."

R2-18

Comment R2-18:

18. Computational-efficiency claims need direct evidence: The manuscript reports approximately 138 GPU-hours for training, but should also report inference time per 3D volume, parameter count, FLOPs, memory use, and direct runtime comparisons with diffusion/flow baselines.

Response:

We thank the reviewer for the valuable suggestion regarding computational efficiency. We agree that reporting training time alone is insufficient to characterize the computational cost of the proposed framework. Therefore, we conducted additional measurements of inference latency, model parameters, and GPU memory usage for 3D PET synthesis.

Using the trained DIReCT++ model on an NVIDIA RTX 4090 GPU with FP32 precision, we evaluated inference performance for volumetric MRI inputs of size $160 \times 192 \times 160$. After 10 warm-up iterations, we performed 50 synchronized timing repetitions for each tracer. The mean inference times were 251.015 ± 0.187 ms for FDG and 251.411 ± 0.256 ms for AV45, where the standard deviations represent repeated measurements on a fixed input. The generator contains approximately 9.93 million parameters, excluding the frozen BiomedCLIP encoders, and its peak PyTorch-allocated GPU memory was approximately 8.29 GiB.

Importantly, the shared DIReCT++ generator produces one tracer-specific PET volume per forward pass. The reported latency therefore reflects a single-tracer generation step with GPU-resident inputs and cached conditioning embeddings, excluding text encoding, image preprocessing, data transfers, and file I/O. We now clarify these measurement conditions in the revised manuscript to ensure accurate interpretation of the computational results.

Regarding FLOPs and direct runtime comparisons with diffusion- and flow-based baselines, we currently lack validated measurements covering the complete network operations under identical hardware and timing protocols. We therefore refrain from estimating speedup factors solely from sampling-step counts and instead report the directly measured efficiency metrics.

We appreciate the reviewer's recommendation. These additional measurements provide direct quantitative evidence of the computational requirements of DIReCT++ for single-step volumetric PET generation and allow us to present its efficiency more precisely and transparently.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Competing Methods / 3D DDIM (p. 19, lines 483–483).
- Implementation Details (p. 22, lines 595–595).

R2-19

Comment R2-19:

19. Reproducibility is a strength but can be improved: Public code, trained models, software versions, and hardware details are valuable. The authors should additionally provide fixed data-split files, random seeds, and a repository commit/version used for the reported experiments.

Response:

We thank the reviewer for recognizing our efforts toward code availability and reproducibility. We agree that providing explicit data splits, random seeds, and code version information would further improve the transparency and reproducibility of our study.

The principal model implementations and relevant training materials have already been made publicly available. We have further organized the experimental configurations and reproducibility records, including fixed subject-level data splits, training and evaluation settings, source-code information, and corresponding file checksums.

For data partitioning, we employed a fixed two-fold cross-validation protocol and preserved the associated split records. Regarding random seeds, we distinguish the seed used for data partitioning from other stochastic processes, such as model initialization and training, and report the settings that can be reliably verified.

We acknowledge that some historical execution records from the original experiments were not fully preserved. Consequently, the exact random states and code versions of every earlier run cannot be reconstructed with certainty. We therefore distinguish the documented experimental configurations from the original historical runs rather than treating them as identical.

Furthermore, the sharing of ADNI subject and imaging identifiers will follow the applicable data-use requirements. We now provide reproducibility information in github repository to the extent permitted by these requirements.

We appreciate the reviewer's recommendation and will improve the documentation of experimental settings and code versions to facilitate reproducibility.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / cohort (p. 17, lines 403–403).
- Materials and Data Preprocessing / evaluation split (p. 18, lines 457–457).
- Data Availability (p. 24, lines 641–641).
- tab:reproducibility (Supplementary Materials, p. 9; combined reference copy p. 33).

- Implementation Details (p. 22, lines 595–595).

Revised Text in Manuscript (Materials and Data Preprocessing / cohort):

"This study leveraged neuroimaging data from 1,857 participants enrolled in the Alzheimer's Disease Neuroimaging Initiative (ADNI), spanning cohorts from ADNI-1 through ADNI-4."

R2-20

Comment R2-20:

20. Clinical translation claims should be moderated: Statements regarding widespread screening, clinically actionable biomarkers, and early intervention exceed the evidence from retrospective ADNI/OASIS cohorts. Prospective, diverse, multi-center clinical validation is needed before such claims are justified.

Response:

We appreciate the reviewer's thoughtful suggestions regarding the clinical translation of our work.

One of the primary motivations of DIReCT++ is to explore the feasibility of synthesizing multi-tracer PET images from routine MRI, potentially alleviating the challenges associated with PET acquisition costs, radiation exposure, and limited accessibility. Our evaluations on the ADNI and OASIS-3 datasets provide preliminary evidence of the framework's PET synthesis performance, while regional uptake analyses and downstream classification on ADNI further support its potential clinical relevance.

Nevertheless, translating these findings from retrospective research cohorts into clinical practice requires more comprehensive evidence. We agree with the reviewer that the current results are insufficient to support the direct application of synthetic PET in routine clinical screening or treatment decision-making. Accordingly, we will moderate statements regarding widespread screening, clinically actionable biomarkers, and early intervention to better distinguish the demonstrated findings from potential future clinical applications. We now also clarify that prospective validation in more diverse, real-world clinical populations remains necessary.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Abstract (p. 1, lines 18–18).
- Introduction / contributions (p. 2, lines 69–69).
- Introduction / evaluation summary (p. 3, lines 76–77).
- Results / framework opening (p. 3, lines 84–84).
- Introduction: clinical access scope (p. 2, lines 52–53).

Revised Text in Manuscript (Introduction / contributions):

"Building upon single-tracer DIReCT [3], DIReCT++ (Domain-Informed ReCTified flow) extends

3D rectified flow to a shared network conditioned on the target PET tracer, enabling FDG- or AV45-PET generation in separate forward passes."

Response to Reviewer 3: Major Comments

R3-1

Comment R3-1:

1. Circularity: the conditioning prompt contains the information that defines the diagnostic labels. This is my principal concern. The clinical prompt encodes "neuropsychological scores" (lines 314-315), and Table S1 lists MMSE, CDR, FAQ, GDS, and NPI-Q. In ADNI, diagnostic group membership is defined largely by MMSE, CDR, and Logical Memory II performance, and the authors state that EMCI/LMCI were assigned using Logical Memory II (lines 358-360). The synthetic PET is then fed to a classifier to predict those same labels. Any gain of "MRI + synthetic PET" over "MRI alone" could therefore reflect clinical scores passed through the generator, not molecular information inferred from anatomy.

The following are needed to resolve this:

The exact prompt template, and a list of which scores are included. Is Logical Memory II or CDR in it? The manuscript is inconsistent: line 95 says "age, gender, clinical scores," lines 314-315 say "age, sex, weight, and neuropsychological scores," and Fig. 1 shows only "Age, Gender, ...". A classifier using the clinical variables alone, and one using MRI + clinical variables, as reference baselines. If MRI + clinical variables matches MRI + synthetic PET, the synthetic PET adds nothing. Synthesis and classification without clinical text (tracer token only), and with shuffled or mismatched prompts. If synthetic PET changes substantially when a CN subject's MRI is paired with an AD patient's scores, the images reflect the prompt rather than the patient.

Response:

We thank the reviewer for the thoughtful comments regarding the potential overlap between clinical conditioning information and diagnostic labels. We acknowledge that cognitive scores are closely associated with diagnostic group membership in ADNI, and therefore agree that their roles in PET synthesis and downstream classification warrant careful consideration.

First, we have verified the exact clinical prompt composition, which includes age, sex, weight, and **five clinical assessments: CDR, MMSE, GDS, FAQ, and NPI-Q**. Logical Memory II is not included in the conditioning prompt. To further evaluate the diagnostic contribution of clinical information, we additionally conducted Clinical-only and MRI+Clinical classification experiments. The results demonstrate that clinical variables themselves provide strong diagnostic discrimination, highlighting the importance of considering their contribution when interpreting downstream classification performance using synthetic PET.

Furthermore, we performed an inference-time cognitive-score removal experiment while keeping the generator weights fixed. Removing cognitive scores resulted in only minor changes in regional errors for both FDG and AV45 PET, with the overall synthesis quality remaining relatively stable. This finding suggests that DIReCT++ can maintain comparable PET synthesis quality even without cognitive-score conditioning. One possible explanation is that the rectified flow framework jointly leverages anatomical information from MRI and adapted tracer-specific guidance, potentially reducing its dependence on cognitive-score conditioning.

We would also like to clarify that the primary objective of DIReCT++ is to synthesize multi-tracer PET by integrating MRI with clinically available complementary information. Cognitive scores provide relevant context regarding patients' cognitive status, which is particularly meaningful when studying fine-grained MCI stages such as EMCI and LMCI. We appreciate the reviewer's important observation and will further clarify the role of clinical conditioning in the revised manuscript, while providing a more cautious interpretation of the downstream classification findings and the potential clinical value of synthetic PET.

The fixed-weight score-removal experiment used the separate E1 Full checkpoint family. It cannot remove information already learned from diagnosis-conditioned filled inputs during training. In the completed classification analysis, Clinical-only and MRI+Clinical AUC exceeded MRI+synthetic dual PET in all three tasks; we do not claim independent incremental diagnostic value of the synthetic PET over those clinical fields.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- tab:supp_adni (Supplementary Materials, p. 1; combined reference copy p. 31).
- tab:clinical_missingness (Supplementary Materials, p. 8; combined reference copy p. 32).
- Materials and Data Preprocessing / clinical inputs (p. 18, lines 422–422).
- Results / framework conditioning (p. 3, lines 89–89).
- eq:composite_conditioning (p. 16, lines 359–359).
- Supplementary conditioning methods and tab:supp_tau_adapter (Supplementary Materials, p. 10; combined reference copy p. 34).
- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).
- Supplementary classification tables and ROC (Supplementary Materials, p. 17; combined reference copy p. 41).
- Abstract (p. 1, lines 18–18).
- Supplementary clinical score sensitivity and demographic stratification (Supplementary Materials, p. 26; combined reference copy p. 50).

Revised Text in Manuscript (Materials and Data Preprocessing / clinical inputs):

"In the prepared clinical inputs, mean Mini-Mental State Examination (MMSE) scores decreased from 29.40 in CN to 20.84 in AD; other clinical scores also varied across diagnostic groups (Table S1)."

R3-2

Comment R3-2:

2. Table S1 values appear implausible and may indicate imputation that encodes the label directly. Several within-group standard deviations are not credible for integer-valued clinical scales. Examples:

EMCI MMSE 27.50 ± 0.50, LMCI MMSE 24.47 ± 0.50, LMCI NPI-Q 4.00 ± 0.03, EMCI GDS 1.01 ± 0.14, and LMCI GDS 3.02 ± 0.28. The LMCI mean MMSE (24.47) is also lower than both MCI-U (26.83) and typical ADNI LMCI values, and the CN weight SD (24.13 kg) is unusually large. The authors should explain how these values were computed and whether missing scores were imputed with group means. If they were, the prompt literally encodes group membership, and the classification results in Fig. 5 would be invalid. Please report missingness per variable and per group, and the imputation strategy.

Response:

We thank the reviewer for the detailed comments regarding the distribution of clinical assessments in Table S1 and the handling of missing values.

Upon reviewing our clinical data preprocessing procedure, we confirmed that some clinical assessment scores were missing from the original dataset. During data preparation, missing scores were simulated based on diagnostic-group information to provide complete clinical conditioning inputs for the model.

We acknowledge that, although this procedure enabled the completion of missing input fields, it may have affected the within-group distributions of clinical scores and indirectly introduced diagnostic-group information into the imputed values. Therefore, the relatively small standard deviations observed for certain clinical assessments in the original Table S1 should not be interpreted as reflecting the natural variability of actual clinical observations. We have now recovered the historical first-record selection and diagnosis-conditioned filling rules. Of 9,285 score cells in 1,857 prepared records, 1,751 were observed and preserved and 7,534 were missing in the selected source records. Group-specific missing counts and rates are reported in the Supplementary Materials. Direct recomputation corrected four Table S1 summaries: EMCI GDSCALE is 1.00 ± 0.00; LMCI GDSCALE is 3.00 ± 0.00, CDR 0.50 ± 0.00, and NPI-Q 4.00 ± 0.00. Clinical-score group-comparison p-values have been removed. CN weight SD (24.13 kg) is reproducible from the source values, but the retained 0–227 kg range includes unresolved validity issues. No participants were removed and no values were replaced or newly imputed.

Regarding the reviewer's concern about potential circularity involving diagnostic information, we agree that it is important to distinguish the contribution of clinical conditioning to PET synthesis from the potential influence of diagnosis-related information on downstream classification. In our previously conducted inference-time sensitivity analysis, removing cognitive scores while keeping the trained generator weights fixed resulted in only minor changes in PET synthesis errors. This finding suggests that image generation does not depend entirely on cognitive-score conditioning. Nevertheless, this analysis alone cannot fully exclude the potential effects of the original missing-value completion strategy.

Accordingly, based on the existing experimental results and cognitive-score sensitivity analysis, we will more clearly delineate the potential implications of diagnostic-group-conditioned completion for the interpretation of PET synthesis and downstream classification results.

We appreciate the reviewer for identifying this important data-processing issue. By clarifying the sources of clinical data, the missing-value handling strategy, and its implications for interpreting the results, we aim to improve the transparency and reliability of our study.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- tab:supp_adni (Supplementary Materials, p. 1; combined reference copy p. 31).
- tab:clinical_missingness (Supplementary Materials, p. 8; combined reference copy p. 32).
- Materials and Data Preprocessing / clinical inputs (p. 18, lines 422–422).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Abstract (p. 1, lines 18–18).

Revised Text in Manuscript (Materials and Data Preprocessing / clinical inputs):
"In the prepared clinical inputs, mean Mini-Mental State Examination (MMSE) scores decreased from 29.40 in CN to 20.84 in AD; other clinical scores also varied across diagnostic groups (Table S1)."

R3-3

Comment R3-3:
3. "Prognostic stratification" is not supported by the analysis. EMCI vs. LMCI is a cross-sectional severity distinction based on a memory cutoff, not a prognostic outcome. The manuscript also describes LMCI as indicating "proximity to dementia conversion" and MCI-U as lacking "clear progression status" (lines 360-362), which conflates the two concepts. To justify the title and the claims in lines 178-181 and 249-250, the authors should perform a genuine prognostic analysis: progressive vs. stable MCI (pMCI/sMCI) over a defined follow-up (e.g., 36 months), with time-to-conversion modeling (Cox regression, C-index). Otherwise, "prognostic" and "stratification" should be removed from the title and claims.

Response:
We thank the reviewer for further highlighting the important distinction between MCI stage discrimination and longitudinal prognostic prediction.

Our original motivation for relating EMCI/LMCI classification to disease progression was based on previous studies examining the clinical characteristics and subsequent conversion risks associated with these two MCI stages (Aisen et al., 2015; Lin et al., 2022; Petersen, 2021). Accordingly, we included EMCI-versus-LMCI classification as a downstream task to investigate the performance of synthetic PET in fine-grained MCI stage stratification.

We agree that EMCI and LMCI in our study were defined according to baseline memory impairment, and that we did not perform progressive-versus-stable MCI (pMCI/sMCI) classification or time-to-conversion analysis based on longitudinal follow-up outcomes. Therefore, the current experiments do not establish the model's ability to predict longitudinal disease progression.

However, we would like to clarify that *stratification* in our study refers to grouping MCI patients according to their current degree of cognitive impairment, rather than stratification according to future disease-conversion risk. To avoid ambiguity, we now remove or revise unsupported prognostic claims throughout the manuscript and clarify that the task concerns MCI stage stratification. We now also revise the descriptions in the Methods section that characterize LMCI as indicating proximity to dementia conversion and MCI-U as lacking clear progression status, ensuring that the group definitions accurately reflect the cross-sectional classification criteria used in our study.

Accordingly, we now revise the manuscript title to **"Precision Synthesis of Multi-Tracer PET via VLM-Modulated Rectified Flow for MCI Stage Stratification"** to clarify the scope of the stratification task.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / diagnostic groups (p. 17, lines 413–413).
- Results / classification and fig:5 (p. 9, lines 196–196).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Abstract (p. 1, lines 18–18).
- Introduction / evaluation summary (p. 3, lines 76–77).
- Title (p. 1).
- PDF metadata.

Revised Text in Manuscript (Materials and Data Preprocessing / diagnostic groups):

"Participants were stratified into five clinically defined diagnostic categories to reflect the full continuum of Alzheimer's disease pathology: Cognitively Normal (CN; $n = 577$), Alzheimer's disease dementia (AD; $n = 317$), and three subtypes of Mild Cognitive Impairment (MCI)."

R3-4

Comment R3-4:

4. Non-significant paired t-tests do not demonstrate equivalence, and group-level agreement does not show subject-level precision. Lines 146-150 infer "high fidelity" from $p > 0.05$ in paired t-tests. Absence of a significant difference is not evidence of equivalence. The central claim of the paper is subject-specific precision (lines 56, 69-73), which group-mean comparisons cannot test: a model that outputs a diagnosis-conditional average image would pass every test in Fig. 4. The authors should provide:

per-subject agreement for regional SUVR (Pearson/Lin's concordance, ICC, Bland-Altman bias and limits of agreement), overall and within each diagnostic group; formal equivalence testing (e.g., TOST) with pre-specified margins; most importantly, amyloid-positivity agreement. Using a standard composite SUVR or Centiloid threshold, report sensitivity, specificity, and Cohen's κ for classifying real $A\beta+$ vs. $A\beta-$ from synthetic AV-45, stratified by CN, MCI, and AD.

The clinically decisive test is whether the model detects amyloid-positive cognitively normal individuals and amyloid-negative MCI. These are exactly the cases where MRI and clinical scores cannot reveal pathology, and where the authors argue the method is most useful (lines 60-61, 217-219).

Response:

We thank the reviewer for emphasizing the importance of subject-level PET uptake agreement and amyloid-positivity assessment. We agree that, beyond evaluating the preservation of disease-related

uptake patterns, quantitative agreement at the individual-subject level is an important aspect of assessing synthetic PET image quality.

In the original study, we used paired regional uptake comparisons and AD-versus-CN group analyses to evaluate the agreement between synthetic and real PET images and to examine whether the synthesized images retained disease-related regional uptake differences. These analyses provided an initial assessment of regional uptake characteristics. However, we agree that the absence of statistically significant mean differences in paired tests alone does not fully characterize subject-level agreement.

To provide a more comprehensive evaluation, we extended our analysis using the preserved subject-matched regional uptake data for both AV45 and FDG PET. Across all 98 regions of interest (ROIs), we calculated Lin's concordance correlation coefficient (CCC), the intraclass correlation coefficient (ICC), mean absolute error (MAE), Bland–Altman bias and 95% limits of agreement, together with mean paired differences and their 95% confidence intervals. These analyses were performed for the overall cohorts and separately within the diagnostic subgroups, allowing us to examine regional agreement, subject-level paired errors, and systematic differences from complementary perspectives.

Across the 98 ROIs in the overall cohorts, the median CCC values were 0.9745 for AV45 and 0.9450 for FDG, while the corresponding median MAEs were 0.0160 and 0.0159, respectively, in normalized intensity units. These results indicate generally strong agreement in regional uptake between synthetic and real PET, while also revealing variations in individual-level errors across brain regions. We now incorporate these additional agreement measures into the revised manuscript and refine the interpretation of the original paired-test results accordingly.

We also appreciate the reviewer's suggestion to evaluate formal equivalence and amyloid-positivity agreement, particularly in amyloid-positive cognitively normal individuals and amyloid-negative patients with MCI. These assessments are highly relevant to the clinical application of synthetic PET. Our present study focuses on MRI-to-PET image synthesis and the reconstruction of regional uptake characteristics. The PET images were processed using per-image min–max intensity normalization, so the resulting regional measurements represent normalized uptake rather than clinically calibrated SUVR or Centiloid values. Consequently, standard amyloid-positivity thresholds cannot be directly applied to these measurements, while formal equivalence testing would require clinically justified equivalence margins.

We now clarify this quantitative scope in the revised manuscript. Future work incorporating PET data with standardized quantitative calibration will enable further evaluation of SUVR or Centiloid agreement and amyloid-positivity classification, including the clinically important atypical cases highlighted by the reviewer.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 6, lines 163–163).
- Methods: regional statistical analysis (p. 20, lines 529–529).
- tab:regional_agreement; tab:regional_effect_sizes (Supplementary Materials, p. 11; combined reference copy p. 35).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).

Revised Text in Manuscript (Results: regional uptake analysis):

"Paired-sample t-tests between synthetic and real images revealed no statistically significant differences in normalized regional uptake across the principal brain regions shown in Fig. 4A, B for the overall cohorts of both tracers ($p > 0.05$)."

R3-5

Comment R3-5:

5. Quantitative definitions: "SUV" vs. SUVR and min-max normalization. All volumes are min-max normalized to [0, 1] per image (lines 388-389). Regional values are then called "SUV" and normalized to cerebellar gray matter or pons (lines 137-138), and plotted as "Mean Uptake" on a 0-1 axis (Fig. 4, S1-S6). Please clarify:

whether ratios were computed on the original PET or on min-max-normalized images (the latter loses quantitative meaning); the reference region for florbetapir (whole cerebellum is the ADNI standard); the correct terminology (SUVR, not SUV).

Please also verify that the FDR correction described in Section 4.5.1 was actually applied to the p-values shown in the figures, and report effect sizes (Cohen's d) for real vs. synthetic group differences. Preserved significance with a much smaller or larger effect size is not "precise preservation."

Response:

We thank the reviewer for the valuable comments regarding the quantitative definition of regional PET uptake, multiple-comparison correction, and effect-size interpretation. We revisited the PET preprocessing and regional statistical analysis procedures used in the original study. We confirmed that the regional uptake values were calculated as the mean intensities within the corresponding brain regions from PET images subjected to per-image min–max normalization. No reference-region ratio normalization using the cerebellum or pons was performed. Therefore, these measurements should not be interpreted as standardized uptake values (SUVs) or standardized uptake value ratios (SUVRs). We now consistently use the term *normalized regional uptake* in the revised manuscript and clarify its calculation and quantitative interpretation.

Regarding multiple-comparison correction, we conducted a systematic re-evaluation using the preserved original regional statistical data. For the AD versus CN comparisons shown in Figure 4C–F, two-sided independent-samples t-tests were performed separately for real and synthetic AV45- and FDG-PET images. Benjamini–Hochberg false discovery rate (FDR) correction was applied across all 98 regions of interest (ROIs) within each tracer and image type, resulting in four comparison families comprising 392 tests. The re-analysis showed that all ten ROIs displayed in Figure 4C/D remained statistically significant in both real and synthetic PET after FDR correction, whereas none of the ten ROIs displayed in Figure 4E/F reached the corrected significance threshold. Thus, all 40 significance/non-significance classifications were consistent with those shown in the original figure.

Furthermore, following the reviewer's suggestion, we calculated Cohen's d for the AD versus CN comparisons in all 20 ROIs displayed in Figure 4C–F, separately for real and synthetic PET, yielding 40 effect-size estimates. Among the ten significant ROIs in Figure 4C/D, the direction of the group differences was consistent between real and synthetic PET. However, the absolute effect sizes decreased in nine of these regions, with a maximum reduction of 0.0771. In Figure 4E/F, two regions exhibited changes in the sign of Cohen's d, but these involved small effect sizes, and neither the real nor synthetic PET comparisons reached statistical significance.

These findings support the preservation of the main disease-related group differences in the representative regions examined, while also revealing variations in their effect-size magnitudes. We acknowledge that the preservation of statistical significance does not establish precise preservation of effect sizes or clinical quantitative equivalence. Accordingly, we now clarify the statistical methods and regional uptake definitions, moderate the relevant interpretations in the revised manuscript, and provide the complete effect-size estimates and FDR-adjusted statistical results in the Supplementary Materials.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 6, lines 149–149).
- Results: regional uptake analysis (p. 9, lines 179–179).
- Methods: regional statistical analysis (p. 21, lines 540–540).
- Methods: regional statistical analysis (p. 21, lines 547–547).
- fig:4 caption (p. 10).
- fig:3 caption (p. 8).
- Supplementary figure 1 (Supplementary Materials, p. 2; combined reference copy p. 25).
- Supplementary figure 2 (Supplementary Materials, p. 3; combined reference copy p. 26).
- Supplementary figure 3 (Supplementary Materials, p. 4; combined reference copy p. 27).
- Supplementary figure 4 (Supplementary Materials, p. 5; combined reference copy p. 28).
- Supplementary figure 5 (Supplementary Materials, p. 6; combined reference copy p. 29).
- Supplementary figure 6 (Supplementary Materials, p. 7; combined reference copy p. 30).
- tab:regional_agreement; tab:regional_effect_sizes (Supplementary Materials, p. 11; combined reference copy p. 35).

Revised Text in Manuscript (Results: regional uptake analysis):

"To assess regional uptake patterns, the corresponding T1-weighted MRI scans were parcellated using SynthSeg [1], and the resulting Desikan–Killiany and aseg ROI masks were applied to spatially aligned real and synthetic PET images."

R3-6

Comment R3-6:

6. Data splitting and possible leakage between generator and classifier. The manuscript states 3-fold cross-validation for synthesis (line 114), two-fold cross-validation (line 391), and 3-fold for classification (line 161). Please clarify:

whether splits are at the subject level (ADNI subjects contribute multiple visits); whether synthetic PET used to test the classifier on a given subject was produced by a generator that did not see that

subject in training; whether generator and classifier folds are nested or aligned; the maximum allowed interval between MRI and PET acquisitions; whether only florbetapir scans were used (ADNI-3/4 also use florbetaben); whether ADNI's pre-processed PET (co-registered, averaged, uniform resolution) was used, and how multi-scanner differences were handled; which subjects (including MCI-U) entered each classification task, with per-task sample sizes.

Response:

We thank the reviewer for the careful assessment of our data splitting strategy, potential information leakage, and image preprocessing procedures. We appreciate the opportunity to clarify these important methodological details.

First, PET synthesis and downstream classification employed two distinct cross-validation protocols. Specifically, PET synthesis was conducted using subject-level two-fold cross-validation. In each fold, the generative model was trained on one subset and used to synthesize PET images for the held-out subjects. The synthetic images from both held-out folds were then combined to form an out-of-fold (OOF) dataset covering the study cohort. The classification analysis retained the existing author-provided synthetic PET inputs under this intended two-fold OOF design. The actual revised classifier partitions and saved outer-test predictions were verified, but the correspondence between these historical synthetic images and the checkpoints used in the reconstructed Full generation comparisons could not be independently established. We therefore report the classification analysis as a separate input-asset family rather than a new downstream validation of those reconstructed generator comparisons. Downstream classification was subsequently performed using an independent three-fold cross-validation scheme. Thus, the two-fold and three-fold protocols correspond to different experimental stages and do not require identical or nested fold assignments.

Second, although ADNI is a longitudinal study, our analysis retained only one imaging record per participant, preventing repeated visits from the same subject from being assigned to different folds. We further verified the subject-level independence of the training and held-out subsets in the two-fold PET generation experiments, as well as the training, validation, and test partitions in the updated three-fold classification experiments. No subject-level overlap was identified within these respective partitions.

Regarding the classification cohorts, the updated experiments included 470 subjects for AD versus CN (165 AD and 305 CN), 741 subjects for MCI versus CN (436 MCI and 305 CN), and 436 subjects for EMCI versus LMCI (287 EMCI and 149 LMCI). The MCI group in these classification experiments comprised subjects explicitly categorized as EMCI or LMCI, while MCI-U subjects were excluded. We now clarify these task-specific sample sizes and selection criteria in the revised manuscript.

Regarding image acquisition and tracer selection, the PET modalities used in this study were AV45 (florbetapir) and FDG, obtained from ADNI, without the inclusion of florbetaben PET. MRI and PET images were paired according to subject identity during dataset preparation. However, the detailed historical records of acquisition-date matching were not fully retained, preventing a reliable reconstruction of the maximum MRI–PET acquisition interval. We therefore refrain from specifying an unverified numerical threshold and acknowledge this limitation in describing the original data selection procedure.

Finally, we appreciate the reviewer's question regarding PET preprocessing and potential multi-scanner variability. We would like to clarify that our study utilized preprocessed PET images downloaded from the official ADNI repository rather than raw PET data. Building upon these

preprocessed images, we implemented a standardized cross-modal preprocessing pipeline, as described in Section 4.3. Specifically, T1-weighted MRI scans underwent skull stripping and histogram matching, followed by rigid registration of each PET image to its corresponding MRI. Both modalities were then affinely normalized to the MNI152 standard space using SPM, with consistent spatial transformations to preserve cross-modal alignment. All images were subsequently standardized to a common spatial dimension of $160 \times 192 \times 160$ voxels at 1-mm isotropic resolution and intensity-normalized to the [0, 1] range using min–max scaling.

These procedures were intended to establish consistent spatial representations and input intensity ranges across subjects and imaging modalities. The use of ADNI-provided preprocessed PET images also helped reduce inconsistencies associated with heterogeneous initial image processing. Nevertheless, we recognize that spatial alignment and intensity normalization do not necessarily eliminate scanner-dependent differences in spatial resolution, reconstruction characteristics, or quantitative response. Accordingly, we now clarify the scope of the preprocessing procedures applied in this study without interpreting them as comprehensive scanner-specific harmonization.

We thank the reviewer again for these constructive comments. In the revised manuscript, we now more clearly distinguish the cross-validation protocols for PET synthesis and downstream classification, specify the subject-level selection criteria and classification sample sizes, and clarify the use of ADNI-preprocessed PET data and the subsequent image registration and normalization procedures. These clarifications will improve the transparency and reproducibility of our experimental methodology.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / clinical inputs (p. 18, lines 422–422).
- Materials and Data Preprocessing / cohort (p. 17, lines 403–403).
- Materials and Data Preprocessing / diagnostic groups (p. 17, lines 413–413).
- Materials and Data Preprocessing / OASIS (p. 18, lines 433–433).
- Materials and Data Preprocessing / preprocessing (p. 18, lines 444–444).
- Materials and Data Preprocessing / evaluation split (p. 18, lines 457–457).
- Data Availability (p. 24, lines 641–641).
- tab:reproducibility (Supplementary Materials, p. 9; combined reference copy p. 33).
- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).

Revised Text in Manuscript (Materials and Data Preprocessing / clinical inputs):

"In the prepared clinical inputs, mean Mini-Mental State Examination (MMSE) scores decreased from 29.40 in CN to 20.84 in AD; other clinical scores also varied across diagnostic groups (Table S1)."

R3-7

Comment R3-7:

7. Classification reporting and claims of significance.

Methods lists ACC, SEN, and SPE (line 459), but Results and Fig. 5 report AUC. No confidence intervals or statistical tests are given, yet the text says "significantly outperforming" (lines 171, 215-216). Please report mean ± SD across folds, 95% CIs, and DeLong tests for AUC differences. EMCI vs. LMCI is imbalanced (291 vs. 152), and specificity is low (~55-75% in Fig. 5C). Please report balanced accuracy. The MRI-only baseline for MCI vs. CN appears near chance (ACC ≈ 51% in Fig. 5B), which is well below published MRI-only performance. This inflates the apparent benefit of synthetic PET. Please justify the classifier and training protocol, or use a stronger MRI baseline.

Response:

We thank the reviewer for the constructive suggestions regarding classification performance reporting, statistical evaluation, and the reliability of the MRI-only baseline.

To address these concerns, we re-evaluated the imaging input configurations using subject-level three-fold cross-validation with a consistent DenseNet121 architecture and training protocol. We now report the fold-wise mean and standard deviation, together with 95% bootstrap confidence intervals, for accuracy (ACC), area under the ROC curve (AUC), sensitivity (SEN), specificity (SPE), and balanced accuracy. We also provide paired bootstrap comparisons, fold-wise DeLong tests, and ROC curves with confidence bands.

For the MCI versus CN task highlighted by the reviewer, the MRI-only baseline achieved an ACC of 67.48% ± 0.23% and an AUC of 71.59% ± 1.80% in the updated evaluation, providing a more reliable reference for comparing different imaging inputs.

To account for class imbalance in the EMCI versus LMCI task, we additionally report balanced accuracy. Compared with MRI alone, combining MRI with synthetic dual-tracer PET yielded a 6.91-percentage-point improvement in balanced accuracy (95% CI: 2.02–11.98), accompanied by improved sensitivity for the minority LMCI class, although specificity was lower. This finding suggests that synthetic PET may provide useful information for the challenging task of fine-grained MCI stage discrimination.

We have revised the classification results and their interpretation accordingly to more accurately characterize the performance of different imaging inputs across the three tasks.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / evaluation split (p. 18, lines 457–457).
- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).
- Supplementary classification tables and ROC (Supplementary Materials, p. 17; combined reference copy p. 41).

Revised Text in Manuscript (Materials and Data Preprocessing / evaluation split):

"PET synthesis used subject-level two-fold cross-validation with fixed, approximately equal partitions (929 and 928 participants; split seed 42)."

R3-8

Comment R3-8:

8. The vision-language component is overstated relative to what is implemented.

The adapted tracer guidance is a fixed constant. Adapt_f(c_f) and Adapt_a(c_a) are pre-computed and fixed (lines 306-307), with a scalar scale and a bias vector. They are therefore equivalent to two learned tracer tokens. Describing this as "domain knowledge encoding" and "prompt learning" overstates its role. An ablation replacing it with a randomly initialized learned tracer embedding is needed. Numeric encoding by CLIP. Subject-specific information comes from a frozen BiomedCLIP text encoder applied to a prompt with numbers. CLIP-style text encoders are known to represent numeric values poorly. Please show that the embedding meaningfully separates, for example, MMSE 29 from 24 (embedding distance vs. score difference), and compare against a simple tabular encoder (an MLP on normalized variables). If the tabular encoder performs as well, the VLM framing is not justified. Image features and constraint logic. BiomedCLIP is a 2D model. Please explain how E[f_f] and E[f_a] were computed from 3D PET (which slices, which aggregation). The constraint sim(Adapt_f, Adapt_a) $\geq \tau$ enforces similarity between tracers, which contradicts the stated goal of "preserving inter-tracer differences" (lines 98-99, 277-278). The value of τ, the optimization method for the constrained problem, and a sensitivity analysis are all missing. Notation. Fig. 1B uses α and β, while Eq. 1 uses α_1, α_2, β_1, β_2.

Response:

We sincerely thank the reviewer for the insightful comments regarding the actual contribution of the vision-language model, the representation of numerical clinical information, and the tracer adaptation mechanism. We agree that the role of the pre-trained vision-language model in DIReCT++ and the significance of different conditioning representations should be described more precisely. In response, we provide the following clarifications based on the methodological design and additional experiments conducted during this revision.

1. Precomputed tracer conditioning versus learnable tracer tokens

We appreciate the reviewer's observation regarding the representation of tracer-specific conditioning information. We agree that the precomputed tracer embeddings are used as fixed tracer-specific tokens during generator training and inference. However, we would like to emphasize that keeping these conditioning vectors fixed during generation does not make their information sources equivalent to those of randomly initialized learnable tokens.

DIReCT++ employs a pre-trained BiomedCLIP model to encode biomedical text descriptions of FDG and AV45. These descriptions not only identify the tracer modalities but also incorporate relevant biomedical information. For example, the FDG description includes cerebral glucose metabolism, neuronal energy demands, and metabolic alterations associated with neurodegenerative diseases. In contrast, the AV45 description characterizes amyloid-beta deposition and its spatial distribution across relevant brain regions. By encoding these descriptions into tracer-specific conditioning vectors, our approach aims to leverage the semantic representations acquired through biomedical image-text pretraining to provide medically informed conditioning for different PET modalities.

Compared with learning tracer embeddings directly from random initialization, the principal distinction of this design lies in the origin of the conditioning representations. The tracer features are constructed from text descriptions with explicit biomedical meanings, rather than being determined solely by trainable parameters optimized for the current generation task. Furthermore, the lightweight affine adaptation mechanism is designed to adjust these tracer-text representations using corresponding PET image features, while a cross-tracer similarity constraint regulates the representational relationship between the two tracers.

Therefore, we consider the primary role of BiomedCLIP in this framework to be the provision of biomedical semantic representations for tracer conditioning and a unified text-based conditioning mechanism for multi-tracer PET synthesis. In the revised manuscript, we now describe this mechanism more precisely, emphasizing the construction and adaptation of pre-trained semantic features rather than suggesting continuous end-to-end updating of the vision-language model during PET generation.

2. Numerical clinical encoding and comparison with TabularMLP

We understand the reviewer's concern regarding the ability of CLIP-based text encoders to represent numerical clinical variables. However, we would like to clarify that the purpose of clinical text conditioning in our study is not merely to transform numerical values such as age, MMSE, and CDR into high-dimensional vectors. Instead, our approach incorporates both the numerical measurements and their corresponding medical interpretations into the conditioning representation.

Specifically, the patient-specific clinical prompts in DIReCT++ contain eight variables: age, sex, weight, CDR, MMSE, GDS, FAQ, and NPI-Q. Importantly, these prompts also include descriptions of the clinical meaning and scoring ranges of the corresponding assessments. For example, MMSE is explicitly described as an assessment of cognitive function, together with its scoring range and interpretation, whereas CDR is associated with the clinical assessment of dementia severity. The remaining clinical assessments are similarly accompanied by descriptions of the functions or symptoms they evaluate.

Through this prompt construction strategy, clinical conditioning is represented not simply as a collection of isolated numerical values, but as structured natural-language descriptions combining patient-specific measurements with relevant medical context. The pre-trained BiomedCLIP text encoder then maps these descriptions into a shared biomedical semantic feature space.

In contrast, TabularMLP directly learns feature mappings from structured clinical variables without incorporating additional natural-language descriptions. Therefore, the distinction between the two approaches extends beyond their network architectures to the representation of clinical information and the prior information used during encoding.

We further investigated whether patient-specific clinical conditioning influences the actual generation process. Our implementation-level analyses confirmed that clinical prompts from different patients produce distinct feature vectors and that these conditioning features participate in generation through cross-attention, producing measurable changes in the synthesized outputs.

To further assess different clinical encoding strategies, we conducted additional experiments using a TabularMLP-based conditioning model. This model processes the same eight structured clinical variables through a trainable multilayer perceptron to produce a 512-dimensional patient-conditioning vector, while retaining the tracer conditioning and the overall generative framework.

The experimental results revealed metric-dependent trade-offs between BiomedCLIP-based text

encoding and TabularMLP-based numerical encoding. The full model achieved higher SSIM for both FDG and AV45, whereas TabularMLP showed advantages in certain error-related metrics. These findings suggest that different conditioning representations may influence structural image similarity and voxel-wise reconstruction errors differently.

We believe that the research value of BiomedCLIP-based text encoding lies in its integration of pre-trained biomedical semantic representations with clinical measurements and their medical interpretations, allowing patient-specific information to participate in PET generation within a clinically contextualized representation space. Compared with relying exclusively on numerical feature learning, this provides an alternative conditioning strategy with a clear methodological motivation.

Furthermore, conventional image-quality metrics, including SSIM, PSNR, MSE, and MAE, primarily assess overall image similarity and reconstruction errors. In contrast, clinical information related to disease status and cognitive function may be more closely associated with regional metabolic distributions and disease-related tracer uptake patterns. Therefore, the value of different encoding strategies should also be evaluated in relation to their effects on clinically relevant imaging characteristics. In future work, we plan to incorporate regional SUV/SUVR-based quantitative analyses and more detailed clinical evaluations to investigate the potential contribution of biomedical semantic conditioning to functional image reconstruction.

3. Three-dimensional PET feature extraction and cross-tracer similarity constraint

Regarding the application of the 2D BiomedCLIP model to 3D PET images, we clarify that, in the affine-adapter feature-extraction procedure, each preprocessed PET volume is represented by 160 two-dimensional slices. Each slice undergoes intensity normalization, resizing, and conversion to a three-channel image before being processed by the frozen BiomedCLIP image encoder to obtain a 512-dimensional feature vector. The slice-level features are subsequently aggregated through average pooling to form a subject-level PET representation. The revised implementation applies the same feature-extraction procedure to both FDG and AV45.

Regarding the cross-tracer similarity constraint, we agree that the one-sided inequality presented in the original manuscript does not accurately represent the implemented optimization objective. The adapter optimization employs a two-sided squared soft penalty with a target similarity of τ, rather than simply encouraging the similarity between the two tracer representations to increase continuously.

Specifically, $\tau = 0.82$ corresponds to the two-decimal approximation of the cosine similarity between the original FDG and AV45 BiomedCLIP text embeddings before adaptation (0.8205). We further conducted representation-level analyses using different τ settings and a configuration in which the cross-tracer similarity penalty was removed. The results demonstrate that this constraint regulates the similarity between tracer representations and produces an observable trade-off between text-image feature alignment and inter-tracer representational separation.

Thus, the purpose of this mechanism is to regulate the relative relationship between the adapted representations, rather than simply maximizing their similarity. We now clarify the implemented optimization objective in the revised manuscript and accurately describe the representation-level characteristics demonstrated by the corresponding experiments.

4. Consistency of notation between Figure 1 and Equation (1)

We thank the reviewer for identifying the inconsistency in the notation of affine transformation

parameters between Figure 1B and Equation (1). We now unify the corresponding symbols in the revised manuscript, explicitly distinguishing the scalar scaling parameters and vector bias parameters of the FDG and AV45 adapters to ensure consistency between the schematic illustration and mathematical formulation.

We now also clarify the distinction between the frozen BiomedCLIP encoders and the independently optimized lightweight affine adapters, avoiding terminology that could incorrectly imply end-to-end fine-tuning of the entire BiomedCLIP backbone.

5. Overall clarification

Taken together, we consider the central role of the vision-language model in DIReCT++ to be the provision of biomedical semantic representations for tracer knowledge and patient-specific clinical information, supporting multi-tracer PET synthesis through a shared conditioning mechanism.

Compared with conditioning approaches based solely on randomly initialized vectors or directly encoded structured numerical variables, BiomedCLIP-based text conditioning offers a distinct representation strategy. On the one hand, tracer conditioning originates from text descriptions containing explicit biomedical knowledge. On the other hand, the patient-specific clinical prompts incorporate numerical measurements together with the medical meanings of the corresponding assessments, allowing clinical information to participate in generation within a shared biomedical text representation space.

Our additional analyses demonstrate that patient-specific conditioning influences the generated outputs, that BiomedCLIP-based and TabularMLP-based encoding strategies exhibit trade-offs across different image-quality metrics, and that the tracer adaptation mechanism regulates the relative relationship between tracer representations at the feature level. These findings provide further insight into the roles of different conditioning representations within the PET synthesis framework.

We sincerely appreciate the reviewer's constructive suggestions. In the revised manuscript, we now further clarify the BiomedCLIP feature-encoding procedure, clinical prompt construction, tracer adaptation mechanism, and mathematical notation. Together with the additional experiments, these revisions will provide a more precise description of the methodological motivation and practical role of biomedical semantic conditioning in DIReCT++.

For clarity, the revised manuscript distinguishes the affine-adapter formulation and representation-level analyses from the reconstructed Full generator, which used original fixed tracer-text vectors. The BiomedCLIP backbone is frozen in both cases. We do not attribute the reconstructed Full generator performance to affine adaptation or claim a controlled generator-level frozen-versus-adapted gain.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / framework conditioning (p. 3, lines 89–89).
- fig:1 caption (p. 4).
- Methods / DIReCT++ overview (p. 14, lines 318–318).
- Domain-Adapted VLM / feature extraction (p. 14, lines 326–326).

- eq:align_opt (p. 15, lines 338–338).
- Rectified Flow / V theta definition (p. 15, lines 350–350).
- eq:composite_conditioning (p. 16, lines 359–359).
- Rectified Flow / cross-attention (p. 16, lines 370–370).
- Supplementary conditioning methods and tab:supp_tau_adapter (Supplementary Materials, p. 10; combined reference copy p. 34).
- Results / reconstructed generation evidence (p. 5, lines 120–120).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Introduction / contributions (p. 2, lines 69–69).

Revised Text in Manuscript (Results / framework conditioning):

"Text guidance incorporates tracer descriptions and patient information (age, sex, weight and clinical scores) encoded by frozen BiomedCLIP encoders and injected into the RF model through cross-attention."

R3-9

Comment R3-9:

9. Ablation study is missing. The manuscript reports no ablation beyond the single-tracer DIReCT comparison. Please add, at minimum:

no text; tracer token only; clinical prompt only; frozen vs. adapted tracer embedding; VLM text encoder vs. tabular encoder; one-step distilled vs. multi-step ODE sampling; joint vs. separate tracer generation.

Section 4.4 describes 3D RF as an ablation "isolating the contribution of BiomedCLIP guidance," but it differs from DIReCT++ in several respects at once.

Response:

We thank the reviewer for the constructive suggestion to strengthen the ablation study. We agree that distinguishing the contributions of text guidance, clinical conditioning, tracer representation adaptation, and sampling strategy is important for a more comprehensive assessment of DIReCT++.

We revisited the existing comparisons and supplemented the analyses of key conditioning components. Because the components serve different functions, we distinguish comparisons involving independently trained models from interventions applied only during inference, rather than treating them as interchangeable forms of ablation.

First, regarding text conditioning, the existing 3D rectified flow (3D RF) baseline does not use the BiomedCLIP-enhanced text guidance. However, it also differs from DIReCT++ in other implementation aspects; hence, the performance difference cannot be attributed exclusively to the text guidance module. Fixed-weight patient-conditioning interventions, including empty or mismatched patient features, demonstrate output sensitivity to the supplied context. They are not a

retrained text-free model or a complete no-text ablation; a nonzero output response alone does not establish clinical benefit.

Second, to evaluate the independent contribution of patient-specific clinical information, we conducted a training-level comparison between the Full and Only-tracer configurations, which retains tracer guidance while excluding the patient clinical condition. We also added a training comparison between BiomedCLIP-based text encoding and an encoder for structured clinical variables to assess the contribution of the VLM representation relative to conventional numerical conditioning. These comparisons address the clinical-conditioning and encoder-design aspects of the requested ablations.

Regarding a clinical-prompt-only configuration, the tracer token in our shared multi-tracer framework not only carries tracer-related information but also specifies the target PET modality. Removing it entirely while retaining only the patient clinical information would leave the network without an explicit distinction between AV45 and FDG generation targets. Therefore, this configuration would not represent a well-defined dual-tracer comparison under the current architecture and task definition.

For one-step versus multi-step ODE sampling, the available validated measurements cover single-step generation only; no matched multi-step quality/latency comparison is claimed. The recovered training code fixes t=0 from epoch 140 and fits the PET-minus-MRI endpoint residual without a separately recovered teacher trajectory. For joint versus separate tracer generation, the original manuscript already includes a single-tracer DIReCT comparison. We now clarify that this evaluates the overall difference between separately trained single-tracer models and the shared multi-tracer framework, rather than isolating a synergistic benefit from joint training alone.

Finally, the contribution of frozen versus adapted tracer embeddings is discussed together with the analysis of tracer representation adaptation. We now avoid attributing a performance gain solely to the adaptation module without an appropriately controlled comparison.

The historical estimate of approximately 138 GPU-hours was not independently verified. The revised computational description therefore uses measured single-step forward latency, parameter count and memory, and does not infer a speedup from step counts or claim a completed equal-protocol FLOP/runtime ranking. We distinguish independently trained model comparisons from fixed-weight inference interventions and feature-level adapter analyses. We now distinguish the purposes and evidential limitations of these experiments in the revised manuscript, without interpreting a baseline comparison as a controlled single-component ablation.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / framework conditioning (p. 3, lines 89–89).
- Results / framework inference (p. 3, lines 106–106).
- fig:1 caption (p. 4).
- eq:composite_conditioning (p. 16, lines 359–359).
- Supplementary conditioning methods and tab:supp_tau_adapter (Supplementary Materials, p. 10; combined reference copy p. 34).
- Results / reconstructed generation evidence (p. 5, lines 120–120).

- Competing Methods / scope (p. 19, lines 470–470).
- Competing Methods / 3D RF (p. 19, lines 486–486).
- Competing Methods / Single (p. 19, lines 488–489).
- Competing Methods / details; Image-Quality Evaluation (p. 19, lines 492–492).
- Supplementary generation tables and cases (Supplementary Materials, p. 12; combined reference copy p. 36).
- Results / classification and fig:5 (p. 9, lines 196–196).
- Methods / classification (p. 21, lines 555–556).
- Multi-Task Learning / objective introduction (p. 16, lines 382–382).
- eq:onestep_generation / learning schedule (p. 17, lines 391–392).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Results: PET synthesis (p. 4).

Revised Text in Manuscript (Results / framework conditioning):

"Text guidance incorporates tracer descriptions and patient information (age, sex, weight and clinical scores) encoded by frozen BiomedCLIP encoders and injected into the RF model through cross-attention."

R3-10

Comment R3-10:

10. Baselines and fairness of comparison.

CycleGAN is an unpaired method applied to a paired task, and Swin-UNet is a segmentation architecture. Neither is a strong MRI-to-PET baseline. Directly relevant MRI-to-PET methods cited in the Introduction (e.g., refs. 11-14) are not compared. Most importantly, the baselines receive no clinical information, while DIReCT++ does. A fair comparison requires at least one strong baseline given the same clinical variables (e.g., 3D RF or a regression U-Net with a tabular conditioning branch). Please also clarify whether SegGuidedDiff was run in 2D or 3D, and how baseline hyperparameters were "optimized to their reported best configurations."

Response:

We thank the reviewer for the constructive comments regarding baseline selection and fairness of comparison. We agree that clarifying the input conditions and task-specific adaptations of different methods is important for an objective evaluation of DIReCT++.

First, regarding the use of clinical information, we would like to clarify that not all baseline methods exclude patient-specific clinical inputs. The original DIReCT incorporated individualized clinical conditioning, and the corresponding conditional diffusion and rectified flow implementations also employed clinical text guidance. Nevertheless, differences remain in their conditioning representations

and network architectures. To provide a more directly comparable reference, we included a clinically conditioned 3D regression U-Net baseline.

This model uses the same MRI inputs, tracer identities, and eight clinical variables as DIReCT++, with identical subject splits and preprocessing protocols. DIReCT++ achieved SSIM values of 0.9211 for FDG and 0.8844 for AV45, compared with 0.9003 and 0.8604, respectively, for the clinically conditioned regression U-Net. These findings further support the image synthesis capability of DIReCT++ under matched clinical input conditions.

Second, regarding the MRI-to-PET methods cited in the Introduction but not included in the quantitative comparisons (Refs. 11–14), we further examined their publicly available resources and experimental protocols. Ref. 11 (JDAM) employs a slice-based evaluation protocol that differs from our 3D volumetric setting; Ref. 12 (Multi-view Diffusion) lacks directly reproducible public code and pretrained weights; Ref. 13 (MRI2PET) provides public code, but independent training and evaluation under our unified data splits and protocols would require additional computational resources and experimental time; and Ref. 14 (UniSyn) does not provide complete public reproduction resources and relies on different datasets and pretraining conditions. Consequently, we could not include these methods in a unified quantitative comparison. We now clarify the coverage of the baseline comparisons in the revised manuscript.

Regarding the existing baselines, we selected representative approaches spanning different medical image synthesis paradigms, including CycleGAN, SwinUNETR, diffusion models, and rectified flow. SwinUNETR was adapted for 3D PET intensity regression, while CycleGAN served as a conventional cross-modal image translation reference. SegGuidedDiff was implemented using a 2D slice-based approach with anatomical segmentation guidance. We now clarify these implementation details and the corresponding task adaptations in the revised manuscript.

Furthermore, we now revise the original statement that all baseline hyperparameters were optimized to their reported best configurations and instead describe the actual training parameters and evaluation settings.

We appreciate the reviewer's suggestions. By incorporating a clinically matched regression baseline, clarifying the reasons for not including certain MRI-to-PET methods, and providing more precise descriptions of the existing baseline implementations, we aim to improve the transparency, fairness, and interpretability of the experimental comparisons.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / generation introduction (p. 4, lines 112–112).
- Results / reconstructed generation evidence (p. 5, lines 120–120).
- Competing Methods / scope (p. 19, lines 470–470).
- Competing Methods / SegGuidedDiff (p. 19, lines 481–481).
- Competing Methods / details; Image-Quality Evaluation (p. 19, lines 492–492).
- Supplementary generation tables and cases (Supplementary Materials, p. 12; combined reference copy p. 36).

Revised Text in Manuscript (Results / generation introduction):

"PET synthesis was evaluated using structural similarity (SSIM), peak signal-to-noise ratio (PSNR), mean squared error (MSE) and mean absolute error (MAE)."

R3-11

Comment R3-11:

11. Image-quality metrics and figures. Radar plots normalized to the best method (Fig. 2A) hide absolute values and variance. Please provide:

a table of SSIM, PSNR, MAE, and MSE (mean ± SD), computed within a brain mask (background inflates PSNR/SSIM); paired significance tests against each baseline; per-tracer OASIS results; voxel-wise absolute error maps for representative and worst-case subjects; subject-level scatter plots of real vs. synthetic composite SUVR, colored by diagnosis and amyloid status; ROC curves with confidence bands.

The examples in Fig. 2B-C and Fig. 3 appear hand-selected; the selection criteria should be stated, or cases should be chosen at the median and 10th percentile of performance.

Response:

We thank the reviewer for these detailed and constructive suggestions regarding image-quality evaluation, statistical analysis, and visualization. We agree that normalized radar plots alone are insufficient to convey the absolute magnitude, inter-subject variability, and spatial distribution of reconstruction errors. Accordingly, we have strengthened the quantitative evaluation and provided additional statistical and visual evidence.

First, regarding absolute image-quality metrics and Figure 2, the normalized radar plots were originally intended to facilitate comparisons across SSIM, PSNR, MSE, and MAE, which have different numerical ranges and optimization directions. We recognize that this presentation does not adequately communicate the absolute performance of each method. We have therefore retained Figure 2 and added a quantitative table to the main manuscript reporting the absolute values of all four metrics for the compared methods on ADNI and OASIS.

We also revisited the original experimental records to assess the availability of subject-level statistics. Although the final aggregate metrics underlying Figure 2 were preserved, some historical subject-level evaluation records and corresponding data-partition information could not be fully recovered. Consequently, it would not be reliable to retrospectively derive standard deviations or paired significance tests for every historical baseline. Given the substantial computational cost of retraining all historical methods, we focused our additional validation on representative key comparisons under a common, fixed two-fold subject-level partition.

Specifically, we re-executed DIReCT++ and evaluated it alongside a clinically conditioned regression 3D U-Net, Swin-UNETR, single-tracer DIReCT, a tabular-MLP conditioning baseline, and a tracer-only conditioning baseline. These experiments retained subject-level evaluation records for 1,598 FDG and 1,365 AV45 PET observations. In the Supplementary Materials, we provide the mean ± standard deviation of SSIM, PSNR, MSE, and MAE, together with paired performance differences, 95% bootstrap confidence intervals, effect sizes, and Holm-adjusted statistical tests. We also clarify the conditioning information and training configurations of the compared methods.

In these comparisons, DIReCT++ achieved improvements over the clinically matched regression 3D U-Net across all four whole-volume metrics for both tracers, supported by paired statistical analyses. Comparisons with the other baselines showed metric-dependent outcomes rather than uniform superiority. We have therefore moderated the corresponding conclusions and avoided attributing differences between independently trained models to individual architectural components without sufficient experimental control. We also distinguish the original multi-method comparison from the subject-matched evaluation rather than combining their results in a single statistical analysis.

Second, regarding brain-mask-based evaluation, we agree that background voxels can influence whole-volume image-quality metrics. We therefore additionally calculated SSIM, PSNR, MSE, and MAE within a predefined mask derived from the frozen SynthSeg segmentation labels. The mask includes cortical and subcortical structures, white matter, and other intracranial regions, including ventricular and cerebrospinal fluid compartments. Its anatomical coverage and the valid-window procedure used for masked SSIM are explicitly described in the Supplementary Materials.

After applying the predefined quality-control exclusion, this evaluation included 1,597 FDG and 1,364 AV45 PET observations. DIReCT++ achieved better mean values for all four masked metrics than both the clinically conditioned regression 3D U-Net and Swin-UNETR for both tracers. All 16 paired comparisons reached statistical significance after Holm correction within their respective predefined testing families. Nevertheless, some standardized effect sizes were small, particularly for AV45 error metrics. We therefore distinguish statistical significance from the magnitude and potential practical relevance of the observed improvements. The complete masked metrics, inter-subject standard deviations, confidence intervals, and paired statistical results are provided in the Supplementary Materials.

Third, regarding case selection and voxel-wise error visualization, the examples in Figures 2 and 3 were randomly selected within their respective diagnostic groups to illustrate qualitative PET synthesis results across different clinical categories. We have clarified this selection criterion in the figure captions.

To provide an additional objective assessment, we have prepared supplementary examples using a predefined, performance-based selection procedure. For each tracer, cases were selected according to subject-level SSIM from the fixed two-fold evaluation, including examples near the median and 10th percentile. The corresponding visualizations show the input MRI, real PET, synthetic PET, and voxel-wise absolute error maps.

We also examined the lowest-SSIM cases and identified a previously documented MRI/segmentation quality-control issue affecting the lowest-ranked subject. We distinguish this data-quality issue from reconstruction errors attributable to the model and do not interpret this case as representative of typical model failure.

Fourth, regarding OASIS evaluation and ROC visualization, we clarify that the external evaluation on OASIS was restricted to AV45 PET. The original absolute evaluation metrics are now presented in the main manuscript, and the interpretation of external generalizability is limited to the evaluated tracer and cohort. These historical results are not combined with the new subject-matched ADNI experiments for paired statistical inference.

For downstream classification, we additionally provide ROC curves with 95% bootstrap confidence bands for the AD versus CN, MCI versus CN, and EMCI versus LMCI tasks, based on the predictions from the three-fold classification evaluation. These supplementary figures illustrate the performance and statistical uncertainty of the evaluated input configurations.

Finally, regarding composite SUVR scatter plots and amyloid-positivity analysis, we acknowledge the clinical relevance of these assessments. However, the PET intensities in the present study underwent per-image min–max normalization, and the resulting regional uptake measurements are not clinically calibrated SUVR or Centiloid values. Applying conventional clinical SUVR thresholds or deriving amyloid-positivity classifications directly from these normalized intensities would therefore risk producing quantitatively unsupported conclusions. We have clarified this limitation and identify standardized PET quantification and amyloid-status agreement as directions for future investigation.

Overall, we have retained the original multi-method visualization while adding absolute quantitative results to the main manuscript and incorporating subject-level statistics, brain-mask-based evaluation, paired comparisons, error visualizations, and ROC confidence bands into the Supplementary Materials. These additions provide a more transparent and reproducible assessment of reconstruction quality while ensuring that the interpretation remains consistent with the available evidence.

The final supplementary examples include the median, 10th-percentile and worst QC-passed subject for each tracer, selected using the saved whole-volume SSIM ranking in the same QC-passed set. The originally flagged MRI/mask failure is not substituted into that set. The original Figure 2 absolute-value table is placed in the main Results; new OOF and masked statistics remain distinct.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- fig:3 caption (p. 8).
- Results / generation introduction (p. 4, lines 112–112).
- Results / generation evaluation scope (p. 5, lines 118–118).
- Results / reconstructed generation evidence (p. 5, lines 120–120).
- Results / external AV45 synthesis (p. 5, lines 141–141).
- fig:2 caption (p. 7).
- Competing Methods / details; Image-Quality Evaluation (p. 19, lines 492–492).
- Supplementary generation tables and cases (Supplementary Materials, p. 12; combined reference copy p. 36).
- Supplementary classification tables and ROC (Supplementary Materials, p. 17; combined reference copy p. 41).

R3-12

Comment R3-12:

12. Determinism, conditional-mean behavior, and uncertainty. Because x_0 is the MRI (not noise) and inference is a single deterministic distilled step (Eq. 7), the model effectively performs conditional regression. For an admittedly ill-posed problem (lines 58-61), this tends to produce a conditional mean, which would smooth out individual pathology. Please:

compare against a plain regression U-Net with identical conditioning; discuss whether the model can represent uncertainty; consider providing uncertainty estimates (e.g., ensembles or stochastic

sampling). For any clinical use, a synthetic amyloid image without a confidence measure is risky.

Also, the claim that rectified flow supports "exact one-step sampling" (lines 338-339) holds only for sufficiently straightened flows. The distillation procedure (teacher steps, loss, schedule after epoch 140) needs a full description.

Response:

We thank the reviewer for the insightful comments regarding deterministic generation, conditional regression, and uncertainty estimation.

We agree that MRI-to-PET synthesis is an ill-posed problem and that deterministic one-step generation cannot fully capture the uncertainty associated with individual molecular pathology.

To address the concern regarding conditional regression, we included a clinically conditioned 3D regression U-Net baseline using the same MRI inputs and clinical variables as DIReCT++. DIReCT++ achieved SSIM values of 0.9211 and 0.8844 for FDG and AV45, respectively, compared with 0.9003 and 0.8604 for the regression baseline. These results demonstrate improved image synthesis performance under matched conditioning settings, although they do not establish that the model captures the full conditional distribution.

Regarding one-step generation, we agree that the distilled prediction represents a learned approximation rather than exact integration of the underlying ODE. As also addressed in our response to Reviewer 2, Comment 13, we now correct this terminology and clarify the distillation procedure. Knowledge distillation begins at epoch 140, as specified in the original implementation.

We also acknowledge that the current deterministic framework does not provide calibrated subject-level uncertainty estimates. Consequently, synthetic PET images should not be interpreted as independent diagnostic evidence without appropriate clinical validation. We will explicitly acknowledge this limitation in the Discussion and identify uncertainty quantification, including ensemble-based or stochastic approaches, as a direction for future research.

We appreciate the reviewer's suggestions, which help clarify both the advantages and limitations of the proposed framework.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / framework inference (p. 3, lines 106–106).
- Results / reconstructed generation evidence (p. 5, lines 120–120).
- Competing Methods / details; Image-Quality Evaluation (p. 19, lines 492–492).
- Supplementary generation tables and cases (Supplementary Materials, p. 12; combined reference copy p. 36).
- eq:onestep_generation / learning schedule (p. 17, lines 391–392).
- Implementation Details (p. 22, lines 595–595).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).

Revised Text in Manuscript (Results / framework inference):

"At inference, the shared network predicts a velocity field for the selected tracer from MRI and the corresponding two-token text context."

R3-13

Comment R3-13:

13. External validation is limited. OASIS-3 covers AV-45 only, and CN and AD only (no MCI), with notable age imbalance (66.3 vs. 72.8 years). Please state which prompt fields were available for OASIS (weight, FAQ, GDS, NPI-Q may be missing) and how gaps were handled; the Supplement mentions "secondary screening" without detail. External validation of FDG synthesis and of the MCI tasks would substantially strengthen generalizability claims.

Response:

We thank the reviewer for the constructive comments regarding the scope of external validation and the clinical data used in OASIS-3.

We agree that our external validation on OASIS-3 was limited to AV45-PET synthesis in cognitively normal (CN) and Alzheimer's disease (AD) participants. These results therefore do not establish cross-dataset generalizability for FDG-PET synthesis or MCI-related tasks. As also addressed in our response to Reviewer 2, Comment 7, we now clarify these limitations and appropriately qualify the corresponding generalizability claims in the revised manuscript.

Regarding clinical conditioning, we now clarify the available clinical variables, participant selection criteria, and handling of missing information in the OASIS-3 cohort. We also acknowledge the age difference between the CN and AD groups, which should be considered when interpreting the external validation results.

We appreciate the reviewer's suggestion and will ensure that our conclusions accurately reflect the scope and limitations of the available external validation evidence.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- tab:clinical_missingness (Supplementary Materials, p. 8; combined reference copy p. 32).
- Materials and Data Preprocessing / OASIS (p. 18, lines 433–433).
- Materials and Data Preprocessing / evaluation split (p. 18, lines 457–457).
- OASIS cohort description (Supplementary Materials, p. 1; combined reference copy p. 24).
- Results / generation evaluation scope (p. 5, lines 118–118).
- Results / external AV45 synthesis (p. 5, lines 141–141).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Abstract (p. 1, lines 18–18).
- Introduction / evaluation summary (p. 3, lines 76–77).

Revised Text in Manuscript (Materials and Data Preprocessing / OASIS):

"External evaluation used the Open Access Series of Imaging Studies (OASIS-3), a longitudinal neuroimaging and clinical cohort focused on normal aging and Alzheimer's disease (`https://www.oasis-brains.org`)."

R3-14

Comment R3-14:

14. Clinical positioning and overstatement. The Discussion makes strong claims: "democratize access," "primary care screening," "qualitative leap," "new paradigm," and identifying candidates "most likely to benefit from disease-modifying therapies" (lines 208-234). Synthetic amyloid images cannot substitute for molecular confirmation, particularly for anti-amyloid therapy eligibility, where a false positive has real consequences. The authors should:

temper these claims; discuss the 2024 revised Alzheimer's Association criteria, which define AD biologically by core biomarkers; compare against, or at least discuss, plasma biomarkers such as p-tau217, which are now a low-cost, radiation-free route to amyloid status. They are mentioned only as a future prompt input (lines 236-237).

Response:

We thank the reviewer for the valuable suggestions regarding the clinical positioning of our study and the interpretation of its potential applications.

We agree that the clinical translation claims in the original manuscript should be moderated. Our study evaluated PET image synthesis, regional uptake patterns, and downstream classification performance using retrospective research cohorts, but did not directly establish the effectiveness of synthetic PET for routine clinical screening or treatment decision-making. Accordingly, we now revise the overstated claims regarding widespread clinical screening, identification of candidates for disease-modifying therapies, and transformative clinical impact. We now also clarify that synthetic PET cannot currently substitute for established molecular biomarker confirmation.

We also appreciate the reviewer's suggestion to contextualize our work within the latest AD diagnostic framework. The 2024 revised Alzheimer's Association criteria further emphasize the biological definition of AD based on core biomarkers and recognize the diagnostic role of blood-based biomarkers, including plasma p-tau217 assays that meet appropriate performance requirements. We have expanded the Discussion to address this updated diagnostic framework, highlight the importance of these biomarkers in assessing AD pathology, and further clarify the current research positioning of DIReCT++.

Relevant references: [4, 5].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Abstract (p. 1, lines 18–18).
- Introduction / evaluation summary (p. 3, lines 76–77).

- Bibliography (p. 52, lines 725–726).
- Bibliography.
- Introduction: biological biomarker scope (p. 2, lines 47–48).
- Introduction: clinical access scope (p. 2, lines 52–53).

Revised Text in Manuscript (Introduction / evaluation summary):

"We evaluate image synthesis on ADNI [6] and retain historical AV45 external results from OASIS-3 [7]."

R3-15

Comment R3-15:

15. Incremental novelty over DIReCT (ref. 15). The main differences from the authors' own IPMI 2025 work appear to be the second tracer, the adapted tracer token, and distillation. The manuscript should state the delta explicitly and show, through ablation, which addition drives the gains.

Response:

We thank the reviewer for highlighting the importance of clarifying the methodological contributions of DIReCT++ beyond the original DIReCT. We agree that the key differences and additional contributions should be more explicitly stated.

The original DIReCT established an MRI-to-PET synthesis framework based on rectified flow and biomedical text guidance. Building upon this foundation, DIReCT++ introduces several extensions. First, it extends single-tracer synthesis to both FDG and AV45 PET through a shared generative network with tracer-specific conditioning. Second, it incorporates a lightweight affine adaptation mechanism that aligns tracer-text embeddings with PET image features while regulating cross-tracer representational similarity, providing medically informed tracer conditioning. Furthermore, the framework employs a learned one-step generation strategy and extends the evaluation to multi-tracer PET synthesis and downstream disease classification tasks.

To further investigate these developments, we conducted additional comparisons and sensitivity analyses examining shared versus separate tracer modeling, clinical conditioning strategies, and tracer representation adaptation. These results provide complementary insights into the roles of different components while also revealing metric-dependent trade-offs in image synthesis performance.

In the revised manuscript, we will more explicitly distinguish the methodological extensions introduced in DIReCT++ from those established in DIReCT and interpret their contributions in light of the available experimental evidence, without attributing the overall performance differences to any single component.

Relevant references: [3].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / framework conditioning (p. 3, lines 89–89).

- Results / framework inference (p. 3, lines 106–106).
- fig:1 caption (p. 4).
- Methods / DIReCT++ overview (p. 14, lines 318–318).
- Results / reconstructed generation evidence (p. 5, lines 120–120).
- Competing Methods / scope (p. 19, lines 470–470).
- Competing Methods / Single (p. 19, lines 488–489).
- Multi-Task Learning / objective introduction (p. 16, lines 382–382).
- Discussion including shared limitations and conclusion (p. 11, lines 230–231).
- Introduction / contributions (p. 2, lines 69–69).
- Results / framework opening (p. 3, lines 84–84).

Revised Text in Manuscript (Results / framework conditioning):

"Text guidance incorporates tracer descriptions and patient information (age, sex, weight and clinical scores) encoded by frozen BiomedCLIP encoders and injected into the RF model through cross-attention."

Response to Reviewer 3: Minor Comments

M1 (Reviewer 3 Minor 1)

Comment M1:

Line 57: "This challenges stems" → "This challenge stems."

Response:

We thank the reviewer for pointing out this grammatical error. We have corrected "This challenges stems" to "This challenge stems" in the revised manuscript.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Introduction / ill-posed inverse problem (p. 2, lines 54–54).

Revised Text in Manuscript (Introduction / ill-posed inverse problem):

"Generative artificial intelligence (AI) offers a transformative pathway to bridge this clinical gap by synthesizing PET biomarkers from routinely acquired MRI [8, 9]."

M2 (Reviewer 3 Minor 2)

Comment M2:

Lines 58-59: "neurodegenration" → "neurodegeneration."

Response:

We thank the reviewer for identifying this typographical error. We have corrected "neurodegenration" to "neurodegeneration" in the revised manuscript.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Introduction / ill-posed inverse problem (p. 2, lines 54–54).

Revised Text in Manuscript (Introduction / ill-posed inverse problem):

"Generative artificial intelligence (AI) offers a transformative pathway to bridge this clinical gap by synthesizing PET biomarkers from routinely acquired MRI [8, 9]."

M3 (Reviewer 3 Minor 3)

Comment M3:

Line 137: "standardized uptake value (SUV) were" → "values (SUVRs) were."

Response:

Thank you for pointing out this grammatical and terminology issue. We agree that the original wording was imprecise. Upon reviewing the regional uptake analysis, we confirmed that the reported measurements represent mean regional intensities extracted from min–max-normalized PET images, rather than standardized uptake values (SUVs) or reference-region–normalized uptake ratios (SUVRs).

Therefore, directly replacing "SUV" with "SUVR" would not accurately reflect the measurements performed in this study. We now correct the grammatical error and use the more appropriate term "normalized regional uptake" in the revised manuscript, while ensuring consistent terminology and units in the corresponding figures.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 6, lines 149–149).

Revised Text in Manuscript (Results: regional uptake analysis):

"To assess regional uptake patterns, the corresponding T1-weighted MRI scans were parcellated using SynthSeg [1], and the resulting Desikan–Killiany and aseg ROI masks were applied to spatially aligned real and synthetic PET images."

M4 (Reviewer 3 Minor 4)

Comment M4:

Line 301: "a learnable velocity field Vθ denotes the network parameters" is garbled; θ and c should be defined separately.

Response:

We thank the reviewer for identifying the unclear notation. We now clarify that V_θ denotes the learnable velocity field and θ its network parameters. We now also use $\mathbf{c}$ consistently for the conditioning context, retaining its component-wise definition in the subsequent formulation.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Rectified Flow / V theta definition (p. 15, lines 350–350).
- eq:onestep_generation / learning schedule (p. 17, lines 391–392).

Revised Text in Manuscript (Rectified Flow / V theta definition):

"The core of DIReCT++ is a continuous generative dynamics framework grounded in the principle of rectified flow, designed to construct a near-optimal, domain-informed trajectory that minimizes sampling complexity while preserving the structural and semantic fidelity of medical image synthesis."

M5 (Reviewer 3 Minor 5)

Comment M5:

Section 4.5.1 describes paired tests for "AD and CN," but Results report CN, MCI, and AD (lines 145-147). Please reconcile.

Response:

We thank the reviewer for identifying this inconsistency between the Methods and Results sections. We have reviewed the regional statistical analyses and clarified the distinction between two complementary types of comparisons.

First, to assess regional uptake agreement between real and synthetic PET, we performed paired two-sided t-tests within the overall cohort and separately within each diagnostic group. The representative results for the overall cohorts are shown in Fig. 4A and B, with additional group-specific statistics provided in the Supplementary Materials.

Second, to evaluate the preservation of disease-related regional differences, we performed independent two-sided Student's t-tests comparing AD and CN subjects separately in real and synthetic PET. These comparisons are illustrated in Fig. 4C–F.

We agree that the original wording did not clearly distinguish these two analyses. Accordingly, we have revised the relevant descriptions in the Results and Methods sections to clarify the populations, comparison types, and corresponding multiple-comparison correction procedures. These revisions ensure consistency between the statistical methodology and reported findings without changing the underlying analyses or results.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 6, lines 157–157).
- Methods: regional statistical analysis (p. 20, lines 529–529).

Revised Text in Manuscript (Results: regional uptake analysis):

"For a comprehensive quantitative evaluation, we performed two complementary regional statistical analyses on the ADNI cohort."

M6 (Reviewer 3 Minor 6)

Comment M6:

Line 427: "Temporoparietal junction" is not a SynthSeg/Desikan-Killiany label. Please specify the ROIs actually used.

Response:

We thank the reviewer for pointing out the inconsistency between the anatomical terminology and the atlas labels used in our analysis. In the original manuscript, we used the general anatomical term "temporoparietal junction," which does not correspond to a distinct ROI label in the adopted atlas and may have caused ambiguity regarding the regions actually analyzed.

To clarify this issue, we re-examined the original ROI statistical analysis and Figure 4 plotting code and confirmed the specific regions used. For FDG-PET, the selected regions included the bilateral hippocampus (Left/Right-Hippocampus), isthmus cingulate cortex (ctx-lh/rh-isthmuscingulate), and

supramarginal gyrus (ctx-lh/rh-supramarginal). For AV45-PET, the selected regions included the bilateral precuneus (ctx-lh/rh-precuneus) and the inferior, middle, superior, and transverse temporal gyri (ctx-lh/rh-inferiortemporal, middletemporal, superiortemporal, and transversetemporal).

Accordingly, we now replace the imprecise or overly broad anatomical descriptions with the corresponding atlas labels and clearly distinguish the full ROI-based statistical analysis from the representative disease-related regions presented in Figure 4. These revisions will ensure that the Methods section accurately reflects the regions used in the analysis and the corresponding figures.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Regional Group-Wise Comparison / ROI definitions (p. 20, lines 521–521).

Revised Text in Manuscript (Regional Group-Wise Comparison / ROI definitions):

"Regional analysis used 98 SynthSeg ROIs defined on T1-weighted MRI."

M7 (Reviewer 3 Minor 7)

Comment M7:

Line 133 vs. line 390: "mri_synthseg (FreeSurfer v7.4.1)" and "SynthSeg [35]" cite different references for the same tool. Please use one version and one citation.

Response:

We thank the reviewer for pointing out the inconsistent nomenclature and citations for the brain parcellation method across the manuscript. We acknowledge that the use of "mri_synthseg (FreeSurfer v7.4.1)" and "SynthSeg" in different sections, together with inconsistent references, may have caused confusion.

To address this issue, we now consistently refer to the ROI parcellation method as **SynthSeg** and standardize its citation throughout the manuscript. We will also clearly distinguish the FreeSurfer-based skull-stripping step from SynthSeg-based brain parcellation and clarify that the ROI masks derived from T1-weighted MRI were used for regional PET analysis. These revisions will ensure a consistent and accurate description of the preprocessing and ROI analysis procedures.

Relevant references: [1].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / preprocessing (p. 18, lines 444–444).
- Results: regional uptake analysis (p. 6, lines 149–149).
- Regional Group-Wise Comparison / ROI definitions (p. 20, lines 521–521).

Revised Text in Manuscript (Materials and Data Preprocessing / preprocessing):

"To ensure precise spatial correspondence between MRI and dual-modality PET images, we implemented a standardized preprocessing pipeline."

M8 (Reviewer 3 Minor 8)

Comment M8:

Refs. 27 and 33 are duplicates (Fischl, FreeSurfer).

Response:

We thank the reviewer for identifying the duplicate reference. We have confirmed that References 27 and 33 refer to the same FreeSurfer publication. We now merge the duplicate entries and standardize the corresponding citations in the revised manuscript to ensure accuracy and consistency.

Relevant references: [10].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Bibliography (p. 52, lines 725–726).
- Bibliography.

M9 (Reviewer 3 Minor 9)

Comment M9:

Ref. 35: "Billot C" should be "Billot B."

Response:

We thank the reviewer for identifying the incorrect author name in Reference 35. We have verified that the first author is Benjamin Billot and will correct “Billot C” to “Billot B” in the revised manuscript.

Relevant references: [1].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Bibliography (p. 52, lines 725–726).
- Bibliography.

M10 (Reviewer 3 Minor 10)

Comment M10:

Ref. 28: author list and title should be verified.

Response:

We thank the reviewer for identifying the inaccuracies in Reference 28. Upon verification, we found that its author list, title, and publication details were inconsistent with the official SynthSeg publication. We now remove the incorrect entry and consistently cite the correct SynthSeg paper by

Billot et al., published in *Medical Image Analysis* in 2023, to ensure the accuracy and consistency of the references.

Relevant references: [1].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Bibliography (p. 52, lines 725–726).
- Bibliography.

M11 (Reviewer 3 Minor 11)

Comment M11:

Ref. 26: author formatting is garbled ("Xingchao Liu Chengyue Gong QL").

Response:

We thank the reviewer for identifying the author formatting error in Reference 26. We have verified that the correct authors are Xingchao Liu, Chengyue Gong, and Qiang Liu. We now correct the author list and update the reference to the officially published ICLR 2023 version to ensure accuracy and consistency.

Relevant references: [11].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Bibliography (p. 52, lines 725–726).
- Bibliography.

M12 (Reviewer 3 Minor 12)

Comment M12:

Ref. 34 (a gray-matter asymmetry protocol) is not an appropriate citation for SPM spatial normalization.

Response:

We thank the reviewer for pointing out the inappropriate citation for SPM. The original reference primarily describes voxel-wise gray matter asymmetry analysis rather than the SPM software used for spatial normalization, and it's our mistake. We now replace it with an authoritative reference for SPM and its associated methods to ensure that the citation accurately reflects the neuroimaging preprocessing tool used in our study.

Relevant references: [12, 13].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Materials and Data Preprocessing / preprocessing (p. 18, lines 444–444).
- Bibliography (p. 52, lines 725–726).
- Bibliography.

Revised Text in Manuscript (Materials and Data Preprocessing / preprocessing):
"To ensure precise spatial correspondence between MRI and dual-modality PET images, we implemented a standardized preprocessing pipeline."

M13 (Reviewer 3 Minor 13)

Comment M13:
Ref. 36 (an ADNI overview) is not an appropriate source for FDG AD-vulnerable regions.

Response:
We thank the reviewer for pointing out the inappropriate citation for AD-vulnerable regions in FDG-PET. We acknowledge that the original reference primarily describes the Alzheimer's Disease Neuroimaging Initiative (ADNI) and does not directly support the regional metabolic alterations discussed in our study.

To address this issue, we now replace the original citation with more relevant FDG-PET studies, including Mosconi et al. on hippocampal hypometabolism, He et al. on AD-related regional metabolic abnormalities, and Strom et al. on metabolic alterations involving the isthmus cingulate region. These references provide a more appropriate scientific basis for the selected disease-related regions.

Relevant references: [14–16].

Action in Manuscript:
The corresponding changes are implemented in the following final manuscript locations:

- Regional Group-Wise Comparison / ROI definitions (p. 20, lines 521–521).
- Bibliography (p. 52, lines 725–726).
- Bibliography.

Revised Text in Manuscript (Regional Group-Wise Comparison / ROI definitions):
"Regional analysis used 98 SynthSeg ROIs defined on T1-weighted MRI."

M14 (Reviewer 3 Minor 14)

Comment M14:
Ref. 25 (MRI-to-CT for spinal segmentation) should be checked as the source for the 3D DDIM baseline.

Response:
We thank the reviewer for pointing out the inappropriate reference for the 3D DDIM baseline.

We re-examined the original experimental code and confirmed that our baseline was not based on the implementation by Graf et al. Instead, it was implemented using the 3D DiffusionModelUNet provided by MONAI GenerativeModels, with DDIMScheduler employed for accelerated sampling. We now remove the inappropriate citation to Graf et al. and replace it with the original DDIM paper by Song et al. and an appropriate reference for MONAI GenerativeModels. We now also clarify the implementation framework and sampling procedure of the 3D conditional diffusion baseline to ensure that the methodological description and references accurately reflect the experimental implementation.

Relevant references: [17, 18].

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results / generation introduction (p. 4, lines 112–112).
- Competing Methods / 3D DDIM (p. 19, lines 483–483).
- Bibliography (p. 52, lines 725–726).
- Bibliography.

Revised Text in Manuscript (Results / generation introduction):

"PET synthesis was evaluated using structural similarity (SSIM), peak signal-to-noise ratio (PSNR), mean squared error (MSE) and mean absolute error (MAE)."

M15 (Reviewer 3 Minor 15)

Comment M15:

Data Availability mentions only "structural MRI and AV-45 PET"; FDG-PET should be added. Please also release the list of ADNI/OASIS subject and image IDs used, to allow exact replication.

Response:

We thank the reviewer for identifying the omission in the Data Availability statement and for emphasizing the importance of data traceability and reproducibility.

We acknowledge that FDG-PET was inadvertently omitted from the original statement. We now correct this omission and clarify the imaging modalities used in each dataset, distinguishing the multi-tracer PET data from ADNI and the AV45-PET data used for external validation on OASIS-3.

Regarding subject and image identifiers, we agree that detailed data selection and splitting information is important for reproducibility. We will organize the data selection criteria, fixed data splits, random seeds, and relevant reproducibility materials for the revised experiments and make them available to the extent permitted by the applicable ADNI and OASIS-3 data use agreements. Any sharing of subject- or image-level identifiers will comply with the corresponding data access and redistribution requirements.

We now also review and correct the data access and licensing information in the Data Availability statement to ensure consistency with the requirements of the respective data providers.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Data Availability (p. 24, lines 641–641).

Revised Text in Manuscript (Data Availability):

"The data used in this study were obtained from two research datasets: the Alzheimer's Disease Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS-3)."

M16 (Reviewer 3 Minor 16)

Comment M16:

Missing hyperparameters: τ, λ_F, λ_A, batch size, U-Net depth and channel widths, cross-attention placement, and parameter count.

Response:

We thank the reviewer for highlighting the insufficient reporting of model architecture details and training hyperparameters. We agree that providing these specifications is essential for methodological transparency and reproducibility.

We have further examined the original experimental notebooks and the configuration used for the revised experiments. The verified loss weights are λ_F = 1.0 and λ_A = 1.0, with a batch size of 1. The generator U-Net consists of four resolution levels with channel widths of (32, 64, 64, 128) and one residual block per level. Cross-attention is configured only at the deepest resolution level, with a conditioning dimension of 512. By instantiating the corresponding U-Net architecture, we verified that it contains 9,925,921 parameters, all of which are trainable. This count refers specifically to the generator U-Net and excludes the VLM and separate adaptation modules.

Regarding the similarity threshold τ, we verified that the revised E1 Full configuration uses $\tau = 0.82$ as a target in a soft penalty rather than as a strictly enforced constraint. We now therefore clarify the distinction between this implementation and the hard-constraint formulation presented in the original manuscript, and revise the mathematical description to accurately reflect the implemented optimization procedure.

In the revised manuscript, we now provide the verified architectural specifications and hyperparameters, clarify the placement of cross-attention, and explicitly define the scope of the reported parameter count. We now also distinguish settings confirmed from the original experimental records from those verified in the reconstructed experiments, ensuring that the reported implementation details are accurate and reproducible.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- eq:align_opt (p. 15, lines 338–338).
- Rectified Flow / cross-attention (p. 16, lines 370–370).
- eq:multitask_loss / definitions (p. 17, lines 387–387).
- Implementation Details (p. 22, lines 595–595).

Revised Text in Manuscript (eq:align_opt):

"where α_1, β_1 are scalar scales and α_2, β_2 are 512-dimensional bias vectors."

M17 (Reviewer 3 Minor 17)

Comment M17:

Fig. 1 legend: "Just Add It" should be replaced with a clear description (e.g., "element-wise addition").

Response:

We thank the reviewer for pointing out the unclear terminology in the legend of Figure 1. We agree that the phrase "Just Add It" is informal and does not adequately describe the underlying mathematical operation.

After reviewing the correspondence between the schematic and the one-step generation formulation, we confirmed that this operation represents element-wise addition of the input MRI and the predicted velocity field to obtain the synthesized PET image.

Accordingly, we now replace "Just Add It" with "Residual Addition" in the revised Figure 1 to provide a more precise and technically appropriate description of the operation.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- fig:1 caption (p. 4).

M18 (Reviewer 3 Minor 18)

Comment M18:

Fig. 4 C-F show only AD vs. CN. Please add MCI comparisons, given the paper's focus on MCI.

Response:

We thank the reviewer for the suggestion to include MCI comparisons in Figure 4C–F. We appreciate the importance of evaluating synthetic PET images in individuals with mild cognitive impairment.

The primary purpose of Figure 4C–F is to investigate whether synthetic PET images preserve representative AD-related regional differences observed in real PET images. We therefore selected AD and CN as contrasting groups to provide a clearer comparison of regional uptake patterns across distinct disease states. Given the clinical and pathological heterogeneity of MCI, whose imaging characteristics may overlap with those of both CN and AD, these panels were not intended to provide an exhaustive comparison across all diagnostic stages.

Instead, our evaluation of MCI focuses on the potential utility of synthetic PET for distinguishing cognitive impairment stages. Specifically, Figure 5 includes dedicated downstream classification tasks for MCI versus CN and EMCI versus LMCI, assessing the contribution of synthetic PET to MCI identification and the differentiation of early- and late-stage MCI. These analyses complement the AD-related regional comparisons presented in Figure 4C–F.

We acknowledge that the original manuscript did not sufficiently clarify this rationale. We now revise the relevant descriptions to explain the purpose of the AD/CN comparisons and the complementary role of the MCI-related classification experiments. We now also clarify that the overall regional analyses in Figure 4A–B included CN, MCI, and AD subjects without separate diagnostic-group stratification.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- Results: regional uptake analysis (p. 6, lines 157–157).
- Results: regional uptake analysis (p. 9, lines 179–179).
- fig:4 caption (p. 10).

Revised Text in Manuscript (Results: regional uptake analysis):

"For a comprehensive quantitative evaluation, we performed two complementary regional statistical analyses on the ADNI cohort."

M19 (Reviewer 3 Minor 19)

Comment M19:

Axis labels "Mean AV45/FDG Uptake" should read "SUVR," with the reference region stated.

Response:

We thank the reviewer for highlighting the need to clarify the quantitative meaning of the regional uptake measurements displayed in Figure 4 and the corresponding supplementary figures. We agree that the axis labels should accurately reflect the normalization procedure to avoid potential misinterpretation of these measurements as clinically quantitative PET uptake indices.

Upon verification, we clarify that the regional uptake measurements in our study do not represent standardized uptake value ratios (SUVRs). During image preprocessing, each PET image was independently normalized to the [0, 1] intensity range using min–max scaling. The mean normalized voxel intensity within each anatomical region was then calculated for regional agreement analyses between real and synthetic PET and for comparisons between diagnostic groups. This procedure did not involve normalization to the cerebellum, pons, or any other anatomical reference region. Consequently, labeling these measurements as either SUV or SUVR would be inaccurate.

To address this concern, we have revised the vertical axis labels in Figure 4 and the corresponding supplementary figures to "Normalized Regional Uptake (a.u.)". We have also clarified in the figure captions and Methods that these values were derived from per-image min–max normalization rather than reference-region normalization.

Furthermore, we have harmonized the relevant terminology throughout the manuscript and clarified in the Discussion that normalized regional uptake measurements cannot be directly interpreted as clinically calibrated SUVR or Centiloid values. Standardized PET quantification and its agreement with clinically established reference measures remain important directions for future investigation.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- fig:4 caption (p. 10).
- fig:3 caption (p. 8).
- Supplementary figure 1 (Supplementary Materials, p. 2; combined reference copy p. 25).
- Supplementary figure 2 (Supplementary Materials, p. 3; combined reference copy p. 26).
- Supplementary figure 3 (Supplementary Materials, p. 4; combined reference copy p. 27).
- Supplementary figure 4 (Supplementary Materials, p. 5; combined reference copy p. 28).
- Supplementary figure 5 (Supplementary Materials, p. 6; combined reference copy p. 29).
- Supplementary figure 6 (Supplementary Materials, p. 7; combined reference copy p. 30).
- fig:2 caption (p. 7).
- Preamble.

M20 (Reviewer 3 Minor 20)

Comment M20:

Abstract: The abstract in the submission system contains raw LaTeX commands (\noindent, \textbf, [18]F), which should be cleaned.

Response:

We thank the reviewer for pointing out the incorrectly displayed LaTeX commands in the abstract within the submission system. We will correct this error when submitting the revised manuscript.

Action in Manuscript:

The submission-system abstract is supplied as plain text without literal LaTeX commands. The authors will use this text when entering the revised submission in Editorial Manager.

M21 (Reviewer 3 Minor 21)

Comment M21:

Terminology: Please use "sex" consistently rather than alternating with "gender."

Response:

We thank the reviewer for pointing out the inconsistent terminology. We now replace "gender" with "sex" in the relevant passage to ensure consistent terminology throughout the manuscript.

Action in Manuscript:

The corresponding changes are implemented in the following final manuscript locations:

- OASIS cohort description (Supplementary Materials, p. 1; combined reference copy p. 24).

- Results / framework conditioning (p. 3, lines 89–89).

Revised Text in Manuscript (Results / framework conditioning):

"Text guidance incorporates tracer descriptions and patient information (age, sex, weight and clinical scores) encoded by frozen BiomedCLIP encoders and injected into the RF model through cross-attention."

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